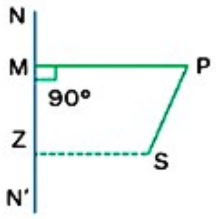
Ellipse

## DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

# Ellipse

In the given figure, S is the focus and NN' is the directrix.




Let P be a point on the ellipse,

then $\frac{PS}{PM} = e, \quad e < 1$ (for ellipse)

Thus, we can find the equation of an ellipse when the coordinates of its focus, equation of the directrix and eccentricity (e) are given.

**Definition**

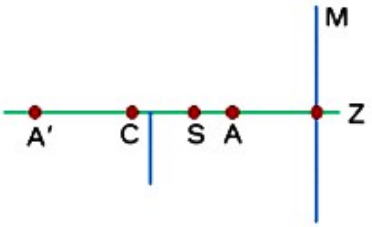
An ellipse is the locus of the point which moves in a plane such that the ratio of its distance from a fixed point (focus) to fixed straight line(directrix) is always constant (called eccentricity).

## STANDARD EQUATION OF AN ELLIPSE :

Let S be the focus & ZM is the directrix of an ellipse. Draw perpendicular from S to the directrix which meet it at Z. A moving point is on the ellipse such that

PS = ePM

then there is point lies on the line SZ and which divide SZ internally at A and externally at A' in the ratio of e : 1 .




therefore SA = eAZ ...(i)

SA' = eA'Z ...(ii)

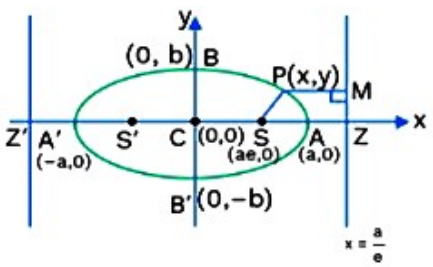
Let AA' = 2a and take C as mid point of AA'

∴ CA=CA' = a

Add (i) and (ii)

SA + SA' = e (AZ + A'Z)

⇒ AA'=e[CZ−CA + CA' + CZ] 2a = 2eCZ



$\Rightarrow \quad CZ = \frac{a}{e}$ ...(iii)

Subtract (ii) and (i), we get

$SA'-SA = e\,(A'Z-AZ)$

$\Rightarrow \quad (CA' + CS) - (CS-CS)$

$= e\,[(CA' + CZ) - (CZ-CA)]$

$\Rightarrow \quad 2CS = 2e\,CA$

$\therefore \; CS = ae$ ...(iv)

Result (iii) and (iv) are independent of axis. consider CZ line as x-axis, C as origin and perpendicular to this line and passes through C is considered as y-axis. Let P(x, y) is a moving point, then by definition of ellipse.

$PS = ePM \Rightarrow (PS)^2 = e^2(PM)^2$

$$\Rightarrow (x - ae)^2 + (y - 0)^2 = e^2\left(\frac{a}{e} - x\right)^2$$

$\Rightarrow (x - ae)^2 + y^2 = (a - ex)^2$

$\Rightarrow x^2 + a^2e^2 - 2axe + y^2 = a^2 + e^2x^2 - 2xae$

$\Rightarrow x^2(1-e^2) + y^2 = a^2\,(1 - e)^2$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2\left(1-e^2\right)} = 1 \quad \text{or} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where $b^2 = a^2\,(1 - e^2)$

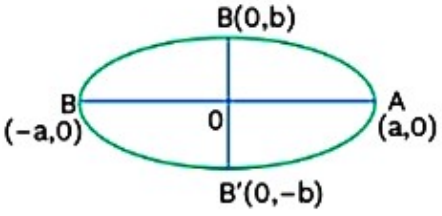
**Tracing of an ellipse :**

Given ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ...(1)

**(i)** If we put y = 0 then we see that ellipse cuts x -axis at (± a, 0).

**(ii)** If we put x = 0 then we see that ellipse cuts y -axis at (0, ± b).

**(iii)** Equation of ellipse does not change when y is replaced by-y. Hence, ellipse is symmetrical about x -axis. (Since equation contain even power of y therefore curve is symmetric about x -axis).



**(iv)** When x is replaced by -x, the equation of curve does not change therefore ellipse is symmetrical about y-axis. (Since equation contain even power of x therefore curve is symmetric about y-axis).

**(v)** From (1) $y = \pm\frac{b}{a}\sqrt{a^2 - x^2}$ , since y is real.

$\therefore a^2 - x^2 \geq 0$ or $-a \leq x \leq a$

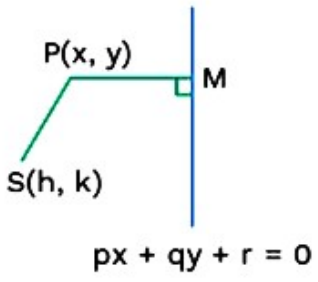
Also from (1), $x = \pm\frac{a}{b}\sqrt{b^2 - y^2}$ , Since x is real. $\therefore b^2 - y^2 \geq 0$ or $-b \leq y \leq b$

Hence ellipse lies entirely between the lines $x = -a$ and $x = a$ and the lines $y = -b$ and $y = b$,

Thus an ellipse is a closed curve. Since curve is symmetrical about both axis, therefore first of all we draw its graph only in the first quadrant and then we will take its image in both axis.

**FACTS ABOUT AN ELLIPSE :**

**(1)** By the symmetry of equation of ellipse, if we take second focus S' (−ae,0) & second directrix $x = -a/e$ and perform same calculation, we get same equation of ellipse, therefore there are two foci & two directrix of an ellipse. The two foci of ellipse are (ae, 0) and (−ae,0) and the two corresponding directrices are lines $x = a/e$ and $x = -a/e$. If focus of the ellipse is taken as (ae,0), then corresponding directrix is $x = a/e$ and if focus is (−ae, 0), then corresponding directrix is $x = -a/e$.




**(2)** If equation of directrix is $px + qy + r = 0$ and focus is (h,k) then its equation will be

$PS^2 = e^2PM^2$

$$(x-h)^2 + (y-k)^2 = e^2 \cdot \left(\frac{px + qy + r}{\sqrt{p^2 + q^2}}\right)^2$$

**(3)** Distance between foci SS' = 2ae and distance between

directrix ZZ' = $2\frac{a}{e}$.

**(4)** Degree of flatness of an ellipse is also called an eccentricity and written as

$$e = \frac{CS}{CA} = \frac{\text{Distance from centre to focus}}{\text{Distance from centre to vertex}}$$

If $e \to 0 \Rightarrow b \to a \Rightarrow$ foci becomes closer and move towards centre and ellipse becomes circle. If $e \to 1 \Rightarrow b \to 0$

$\Rightarrow$ ellipse get thinner and thinner

**(5)** Two ellipse are said to be similar if they have same eccentricity.

**(6)** Distance of focus from the extremity of minor axis is equal to 'a' because

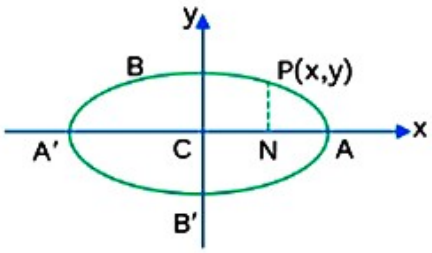
$a^2 e^2 + b^2 = a^2$

**(7)** Let P(x,y) be any point on the ellipse.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \therefore \quad \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$\Rightarrow \frac{y^2}{b^2} = \frac{(a-x)(a+x)}{a^2} \Rightarrow \frac{PN^2}{b^2} = \frac{AN \cdot A'N}{a^2}$$

$$\Rightarrow \frac{PN^2}{AN \cdot A'N} = \frac{b^2}{a^2}$$




**(8)** By definition of ellipse, the distance of any point on the ellipse from focus is equal to e times (the distance of point from the corresponding directrix).

**Let the equation of the ellipse be**

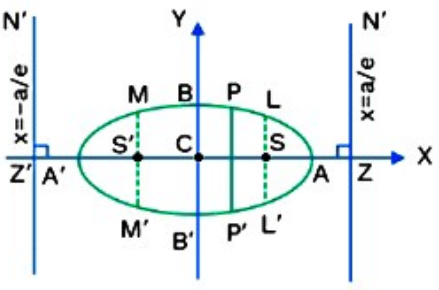
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \ldots(1)$$

**1. Centre:**

In the figure, C is the centre of the ellipse. All chords passing through C are called diameter and bisected at C.

**2. Foci :**

S and S' are the two foci of the ellipse and their coordinates are (ae, 0) and (–ae, 0) respectively. The line containing two foci are



called the focal axis and the distance between S and S' the focal length.

**3. Directrices :**

ZN and ZN' are the two directrices of the ellipse and their equations are $x = \frac{a}{e}$ and $x = \frac{-a}{e}$ respectively. Here Z and Z' are called foot of directrix.

**4. Axes:**

The line segments A'A and B'B are called the major and minor axes respectively of the ellipse. The point of intersection of major and minor axis is called centre of the ellipse. Major and minor axis together are called principal axis of ellipse. Here Semi-Major axis are CA = CA' = a; and Semi-minor axis are CB = CB' = b

**5. Vertex:**

The points where major axis meet the ellipse is called its vertices. In the given figure, A' and A are the vertices of the ellipse.

**6. Ordinate and double ordinates:**

Let P be a point on the ellipse. From P we draw PM perpendicular to major axis of the ellipse. Produce PM to meet the ellipse at P'. then PM is called an ordinate and PMP' is called the double ordinate of the point P. It is also defined as any chord perpendicular to major axis is called its double ordinate.

**7. Latus rectum :**

When double ordinate passes through focus then it is called the Latus rectum.

Let L'L = 2k, then LS = k so L = (ae, k).

Here LL' and MM' are called latus rectum.

Since L (ae,k) lies on the ellipse (1),

therefore $\frac{a^2e^2}{a^2} + \frac{k^2}{b^2} = 1$ or $\frac{k^2}{b^2} = 1 - e^2$

or $k^2 = b^2\left(1 - e^2\right) = b^2 \cdot \frac{b^2}{a^2} = \frac{b^4}{a^2}$ $[\because\ b^2 = a^2\,(1 - e^2)]$

$\therefore \quad k = \frac{b^2}{a}$

Hence length of semi latus rectum $= \frac{b^2}{a} = MS'$

i.e. length of the latus rectum LL' or

$MM' = \frac{2b^2}{a} = \frac{(\text{minor axis})^2}{\text{major axis}}$

$= 2a(1 - e^2)$

= distance from focus to the corresponding directrix

And the end points of latus rectum are

$$L\left(ae, \frac{b^2}{a}\right), L'\left(ae, \frac{-b^2}{a}\right), M\left(-ae, \frac{b^2}{a}\right) \& M'\left(-ae, -\frac{b^2}{a}\right)$$

**8. Focal chord:**

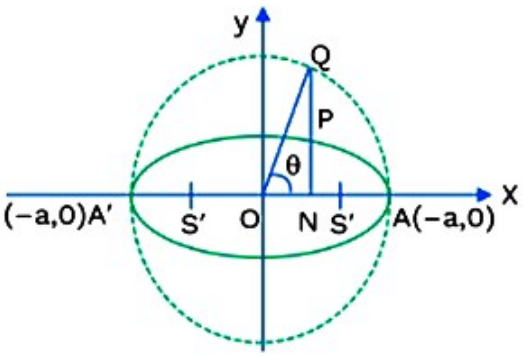
A chord of the ellipse passing through its focus is called a **focal chord.**

## AUXILIARY CIRCLE AND ECCENTRIC ANGLE :

A circle described on major axis as diameter is called the auxiliary circle of the given ellipse & its equation is

$x^2 + y^2 = a^2$ ...(1)

and given ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ...(2)




Let Q be a point on the auxiliary circle $x^2 + y^2 = a^2$ then line through Q and perpendicular to x -axis meet the ellipse at P then P and Q are called the corresponding points on the ellipse & the auxiliary circle respectively. Here $\angle QOA = \theta$ is called the eccentric angle of the point P on the ellipse $(0 \le \theta < 2\pi)$.

Since Q lie on the circle therefore Q $(a\cos\theta, a\sin\theta)$

So coordinate of $P(a\cos\theta, y)$, which satisfy the equation of ellipse.

$$\therefore \frac{a^2\cos^2\theta}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow y = b\sin\theta$$

$\therefore$ coordinate of P will be $(a\cos\theta, b\sin\theta)$ and this is called parametric equation of ellipse.

The equations $x = a\cos\theta$ and $y = b\sin\theta$ together represent the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

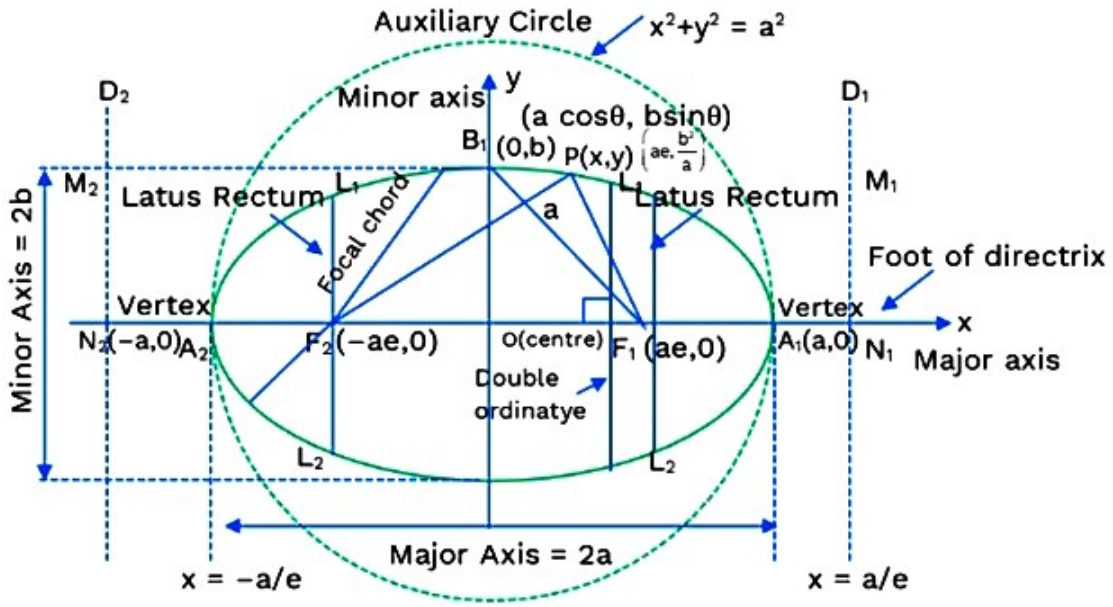
We observe that

$$\frac{\ell(PN)}{\ell(QN)} = \frac{b}{a} = \frac{\text{Semi minor axis}}{\text{Semi major axis}}$$

**Definition**

"If from each point of a circle perpendiculars are drawn upon a fixed diameter then the locus of the points dividing these perpendiculars in a given ratio is an ellipse of which the given circle is the auxiliary circle". This is another definition of ellipse.

## ELLIPSE AT A GLANCE :



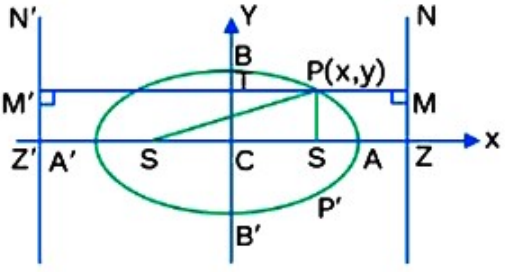

### Focal Distance of a point:

Let P(x,y) be any point on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \ldots(1)$$

Then by definition of ellipse,

$$SP = ePM = e(MT - PT) = e\left(\frac{a}{e} - x\right) = a - ex$$

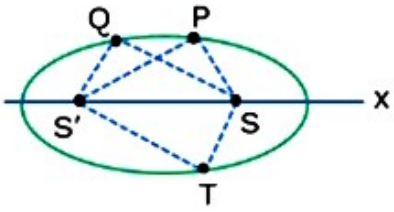


and

$$S'P = ePM' = e\left(M'T + PT\right) = e\left(\frac{a}{e} + x\right) = a + ex$$

Hence SP + S'P = 2a

Because of the above property, ellipse is also defined as the locus of a point which moves in a plane such that the sum of its distance from two fixed points (called foci) is a constant (Length of major axis). This definition is called the physical definition of the ellipse.



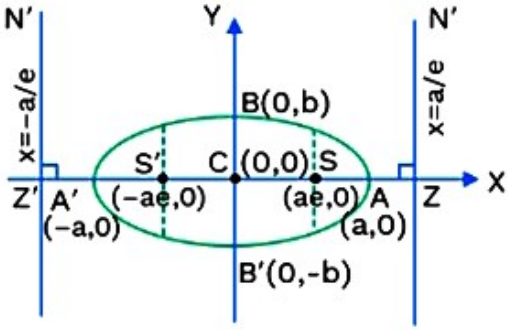

Hence PS +PS' = QS + QS' = TS +TS' = length of major axis

## TWO STANDARD FORMS OF ELLIPSE :

There are two standard forms of ellipse with centre at the origin and axes along coordinate axes. The foci of the ellipse are either on the x-axis or on the y-axis.

### 1. Major axis along x-axis :

The equation of this type of ellipse is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ , where a > b > 0 and $b = a\sqrt{1 - e^2}$ .




**For this ellipse :**

(i) Major axis is 2a

(ii) Minor axis is 2b.

(iii) Centre is (0, 0)

(iv) Vertices are (±a, 0)

(v) Foci are (± ae, 0)

(vi) Equations of directrices are $x = \pm \frac{a}{e}$

(vii) Equation of major axis is y = 0

(viii) Equation of minor axis is x = 0

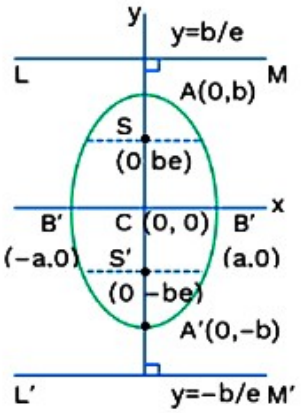
(ix) length of latus rectum = $\frac{2b^2}{a}$

(x) Extremity of latus rectum is $\left(\pm ae, \pm \frac{b^2}{a}\right)$

(xi) Area of ellipse = πab

## 2. Major axis along y-axis :

The equation of this type of ellipse is of the form $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ . where $0 < a < b$ and $a = b\sqrt{1-e^2}$ .




**For this ellipse :**

(i) Major axis is 2b

(ii) Minor axis is 2a

(iii) Centre is (0,0)

(iv) Vertices are (0, ± b)

(v) Foci are (0, ± be)

(vi) Equation of directrices are $y = \pm\frac{b}{e}$

(vii) Equation of major axis is $x = 0$

(ix) Length of latus rectum = $\frac{2a^2}{b}$

(x) Extremity of latus rectum is $\left(\pm\frac{a^2}{b}, \pm be\right)$

(xi) Area of ellipse = $\pi ab$

### COMPARISON CHART BETWEEN STANDARD ELLIPSE :

| Basic Elements | $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ | |
|---|---|---|
| | a> b | a < b |
| Centre | (0,0) | (0, 0) |
| Vertex | (± a, 0) | (0, ±b) |
| Length of major axis | 2a | 2b |
| Length of minor axis | 2b | 2a |
| Foci | (±ae, 0) | (0, ±be) |
| Equation of directrix | $x = \pm\frac{a}{e}$ | $y = \pm\frac{b}{e}$ |
| Relation among a and b | $b^2 = a^2(1-e^2)$ | $a^2 = b^2(1-e^2)$ |
| Equation of major axis | y = 0 | x = 0 |
| Equation of minor axis | x = 0 | y = 0 |
| Length of latus rectum | $\frac{2b^2}{a}$ | $\frac{2a^2}{b}$ |

| | | |
|---|---|---|
| End of latus rectum | $\left(\pm ae, \pm \frac{b^2}{a}\right)$ | $\left(\pm \frac{a^2}{b}, \pm be\right)$ |
| Focal distance of $P(x_1, y_1)$ | $a \pm ex_1$ | $b \pm ey_1$ |
| SP + SP' | 2a | 2b |
| Distance between foci | 2ae | 2be |
| Distance between directrix | $\frac{2a}{e}$ | $\frac{2b}{e}$ |
| Parametric equation | $(a\cos\theta, b\sin\theta)$ $(0 < \theta < 2\pi)$ | $(a\cos\theta, b\sin\theta)$ |

If the equation of the ellipse is given as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and nothing is mentioned then the rule is to assume that a > b.

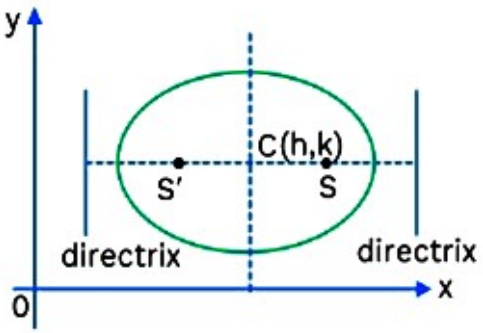
**To find the various Parameter of an ellipse :**

Equation of an ellipse whose axis are parallel to coordinate axis and its centre is (h, k). the foci of the ellipse are either on x-axis or on the y-axis.

**(I) Major axis parallel to x-axis :**

Here the equation of ellipse is of the form $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ , where a > b and $b^2 = a^2(1-e^2)$

Here the equation of ellipse is of the form $\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1$ , where X = x –h and Y = y–k.




(1) Equation of major axis is Y = 0,
i.e., y – k = 0
Equation of minor axis is X = 0,
i.e., x – h = 0

(2) Coordinate of centre of the ellipse are given by X = 0 and Y = 0 i.e, x – h = 0 and y–k = 0
∴ Centre of the ellipse is (h,k)

**(3)** Coordinate of foci of the ellipse are given by
$X = \pm ae, Y = 0$ i.e., $x - h = \pm ae$
and $y - k = 0$
$\therefore$ Hence foci of the ellipse are $(h \pm ae, k)$

**(4)** Equation of the directrices of the ellipse are $X = \pm \frac{a}{e}$, i.e., $x - h = \pm \frac{a}{e}$.
Thus directrices are $x = h \pm \frac{a}{e}$

**(5)** Coordinate of ends of latus rectum are given by $X = \pm ae, Y = \pm \frac{b^2}{a}$ i.e. $x-h = \pm ae$, $y-k = \pm \frac{b^2}{a}$ Therefore ends of latus rectum are given by $\left(h \pm ae, k \pm \frac{b^2}{a}\right)$

(6) Coordinate of vertices of the ellipse are given by $X = \pm a, Y = 0$ i.e., $x-h = \pm a$, $y - k = 0$. Hence vertices are $(h \pm a, k)$

**(II) Major axis parallel to y-axis :**

Here the equation of ellipse is of the form $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, where $a < b$ and $a^2 = b^2 (1 - e^2)$

Equation (1) is of the form $\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1$, where $X = x - h$ and $Y = y - k$

**(1)** Equation of major axis is $X = 0$,
i.e., $x-h = 0$
Equation of minor axis is $Y = 0$,
i.e., $y- k = 0$

**(2)** Coordinate of centre of the ellipse are given by $X=0$ and $Y=0$
$\Rightarrow x-h=0$ and $y-k=0$
$\therefore$ Centre of the ellipse is $(h,k)$

**(3)** Coordinate of foci of the ellipse are given by X = 0, Y = ± be

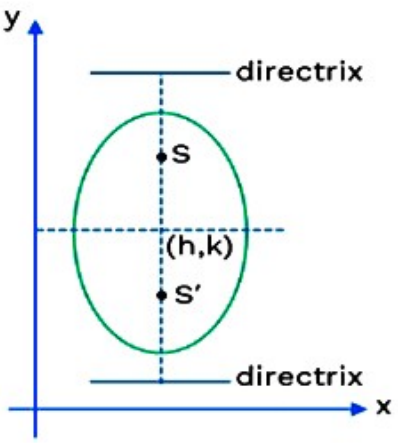
x−h = 0 and y−k = ±be

x = h and y = k ± be

∴ Foci are (h, k ± be)

**(4)** Equation of the directrices of the ellipse are $Y = \pm \frac{b}{e}$ ,i.e,. $y-k = \pm \frac{b}{e}$

Thus directrices are $y = k \pm \frac{b}{e}$.




**(5)** Coordinate of ends of latus rectum are given by

$X = \pm\frac{a^2}{b}$ & $Y = \pm ae$,

i.e. $x - h = \pm\frac{a^2}{b}$ and $y - k = \pm be$,

or $x = h \pm \frac{a^2}{b}$ and $y = k \pm be$

∴ Coordinates of ends of latus rectum are given by $\left(h \pm \frac{a^2}{b}, k \pm be\right)$

**(6)** Coordinates of vertices of the ellipse is given by X = 0 and Y = ± b i.e., x − h=0 and y − k = ± b therefore coordinates of vertex are (h, k ± b)

| Basic Elements | | $\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$ | |
|---|---|---|---|
| | | a > b | a < b |
| 1. | Length of major axis | 2a | 2b |
| | Length of minor axis | 2b | 2a |
| 2. | Equation of major axis<br>Equation of minor axis | y − k = 0<br>x − h = 0 | x − h = 0<br>y − k = 0 |
| 3. | Centre of ellipse | (h, k) | (h, k) |
| 4. | Eccentricity | $e = \sqrt{1-\frac{b^2}{a^2}}$ | $e = \sqrt{1-\frac{a^2}{b^2}}$ |

| | | | |
|---|---|---|---|
| 5. | Foci | $(h \pm ae, k)$ | $(h, k \pm be)$ |
| 6. | Equation of directrix | $x = h \pm \frac{a}{e}$ | $y = k \pm \frac{b}{e}$ |
| 7. | Extremities of latus Rectum | $\left(h \pm ae, k \pm \frac{b^2}{a}\right)$ | $\left(h \pm \frac{a^2}{b}, k \pm be\right)$ |
| 8. | Vertices of an ellipse | (h ± a,k) | (h, k ± b) |
| 9. | Length of latus rectum | $\frac{2b^2}{a}$ | $\frac{2a^2}{b}$ |

**Equation of chord of an Ellipse:**

Equation of a chord of an ellipse joining two points P(α) and Q(β) on it is equal to

$$\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$$

(use formula of line joining points P(acosα, bsinα) and Q(acosβ, bsinβ) ) If this particular chord passes through (d,0) then we have

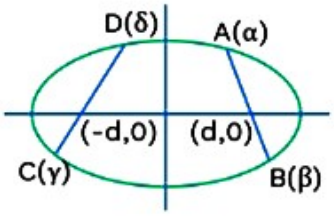
$$\frac{d}{a}\cos\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right);\frac{\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}=\frac{a}{d}$$

Using compodendo and dividendo rule

$$\frac{\cos\left(\frac{\alpha+\beta}{2}\right)-\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)+\cos\left(\frac{\alpha-\beta}{2}\right)}=\frac{a-d}{a+d}$$

or $$-\frac{2\sin\alpha/2\sin\beta/2}{2\cos\alpha/2\cos\beta/2}=\frac{a-d}{a+d}$$

i.e. $$\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{d-a}{d+a}$$




**Point to Remember!!!**

If d = ae i.e. PQ is a focal chord then $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{e-1}{e+1}$

**Q.** **Find the equation of the ellipse (referred to its axis as the x-axis and y-axis) whose foci are (±2, 0) and eccentricity = $\frac{1}{2}$ .**

**Sol.** Let $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ represent ellipse where $e=\frac{1}{2}$

given $(\pm ae, 0) = (\pm 2, 0)$

$\Rightarrow$ $ae = 2 \Rightarrow a = 4 \Rightarrow b^2 = a^2(1-e^2)$ gives $b^2 = 12$

$\therefore$ Equation of ellipse is $\frac{x^2}{16}+\frac{y^2}{12}=1$ .

**Q.** **(a) Find the lengths of the axes, the eccentricity, the coordinates of foci, equations of the directrices and the lengths of latus rectum of the ellipse. Also find the areas enclosed by the curve. $25x^2 + 36y^2 = 900$**

**(b) Find the length and equation of major and minor axes, centre, eccentricity, foci, equation of directrices, vertices and length of the following ellipses**

**(i) $16x^2 + y^2 = 16$**

**(ii) $2x^2 + 3y^2 - 4x - 12y + 13 = 0$.**

**(iii) $4x^2 + y^2 - 8x + 2y + 1 = 0$.**

**Sol.** (a) The equation of the ellipse is

$25x^2 + 36y^2 = 900 \Rightarrow \frac{x^2}{36}+\frac{y^2}{25}=1$

Comparing with $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, we get

$a^2 = 36, b^2 = 25 \Rightarrow a = 6, b = 5$

$\because$ $a > b$, so it is a horizontal ellipse with axis along the X-axis and the centre at the origin.

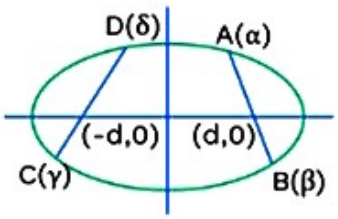
Thus,

Length of major axis = 2a = 12 unit

Length of minor axis = 2b = 10 unit

If e is the eccentricity, then

$b^2 = a^2\left(1-e^2\right)$




$\Rightarrow 25 = 36\left(1-e^2\right) \Rightarrow e = \frac{\sqrt{11}}{6}$

$\therefore$ $ae = 6\cdot\frac{\sqrt{11}}{6} = \sqrt{11}$

The coordinates of foci are (ae, 0) and (– ae, 0)

i. e., $(\sqrt{11}, 0)$ and $(-\sqrt{11}, 0)$

Also, $\frac{a}{e} = \frac{6}{\frac{\sqrt{11}}{6}} = \frac{36}{\sqrt{11}}$

The equations of the directrices are

$x = \frac{a}{e}$ and $x = -\frac{a}{e}$,

i.e., $x = \frac{36}{\sqrt{11}}$ and $x = -\frac{36}{\sqrt{11}}$

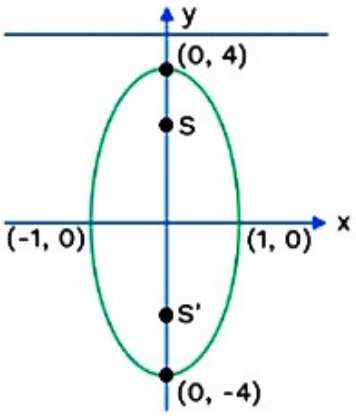
$\Rightarrow \sqrt{11}x - 36 = 0$ and $\sqrt{11}x + 36 = 0$

Length of the latus rectum $= \frac{2b^2}{a} = \frac{2 \cdot 25}{6} = \frac{25}{3}$ unit

Area of the ellipse = $\pi ab = \pi\ 6\ .\ 5 = 30\pi$ sq. unit

**(b) (i)** Given equation of the ellipse is

$16x^2 + y^2 = 16$ or $\frac{x^2}{1^2} + \frac{y^2}{4^2} = 1$ ... (1)




Here a = 1, b = 4 and a < b
Length of major axis = 2b = 8
Length of minor axis = 2a = 2
Equation of major axis is x = 0
Equation of minor axis is y = 0
Coordinates of centre are (0, 0)
Eccentricity of the ellipse,

$e = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{1}{16}} = \frac{\sqrt{15}}{4}$

Coordinate of foci are given by

$y = \pm$ be, $x = 0$ i.e., $y = \pm\sqrt{15}, x = 0$

Hence foci are $(0, \pm\sqrt{15})$

Equation of directrices are $y = \pm\frac{b}{e}$ or $y = \pm\frac{16}{\sqrt{15}}$

Coordinates of vertices are given by
$y = \pm b$ and $x = 0$ i.e., $y = \pm 4$,
$x = 0$
Hence vertices are $(0, \pm 4)$.

**(ii)** The given equation does met contain the term of xy, so its axes are parallel to the coordinates axes. We have

$$2x^2 + 3y^2 - 4x - 12y + 12 = 0$$

or $2\left(x^2 - 2x\right) + 3\left(y^2 - 4y\right) + 13 = 0$

or

$$2\left(x^2 - 2x + 1\right) + 3\left(y^2 - 4y + 4\right) - 14 + 13 = 0$$

or $2(x-1)^2 + 3(y-2)^2 = 1$

or $\dfrac{(x-1)^2}{\frac{1}{2}} + \dfrac{(y-2)^2}{\frac{1}{3}} = 1$ ...(1)

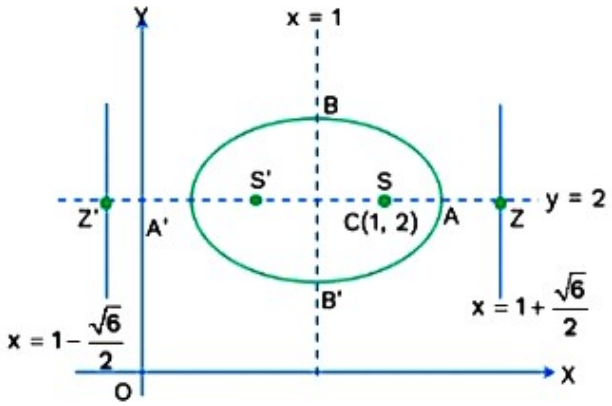
Shifting the origin to the point (1, 2) equation (1) becomes

$$\frac{X^2}{\frac{1}{2}} + \frac{Y^2}{\frac{1}{3}} = 1$$

Comparing with $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, we get

$$a^2 = \frac{1}{2}, b^2 = \frac{1}{3} \Rightarrow a = \frac{1}{\sqrt{2}} \text{ and } b = \frac{1}{\sqrt{3}}$$

$\because$ $a > b$, so the major axis is parallel to x - axis.




Length of major axis

$$= 2a = 2 \cdot \frac{1}{\sqrt{2}} = \sqrt{2} \text{ unit}$$

Length of minor axis

$$= 2b = 2 \cdot \frac{1}{\sqrt{3}} = \frac{2}{3}\sqrt{3} \text{ unit}$$

Length of latus rectum $= \frac{2b^2}{a} = 2\cdot\frac{1}{3}\cdot\sqrt{2} = \frac{2\sqrt{2}}{3}$ unit

Eccentricity, $e = \sqrt{1-\frac{b^2}{a^2}} = \sqrt{1-\frac{2}{3}} = \frac{1}{\sqrt{3}}$

Coordinates of the centre and the foci and the equations of the directrices are given in the following table:

| | X, Y System | x, y System |
|---|---|---|
| Coordinates of centre | (0, 0), i.e, X = 0, Y = 0 | $x - 1 = 0$ and $y - 2 = 0$<br>$\Rightarrow x = 1, y = 2$<br>$\therefore$ Centre is (1, 2) |
| Coordinates of force | (ae, 0) and (−ae, 0) or $\left(\frac{1}{\sqrt{6}}, 0\right)$ and $\left(-\frac{1}{\sqrt{6}}, 0\right)$ | $x - 1 = \frac{1}{\sqrt{6}}, y - 2 = 0$ and $x - 1 = -\frac{1}{\sqrt{6}}, y - 2 = 0$<br>$\Rightarrow x = 1 + \frac{1}{\sqrt{6}}, y = 2$ and $x = 1 - \frac{1}{\sqrt{6}}, y = 2$<br>$\therefore$ Foci are $\left(\frac{\sqrt{6}+1}{\sqrt{6}}, 2\right)$ and $\left(\frac{\sqrt{6}-1}{\sqrt{6}}, 2\right)$ |
| Equations of directrices | $X = \frac{a}{e}$ and $X = -\frac{a}{e}$ or $X = \frac{\sqrt{6}}{2}$ and $X = -\frac{\sqrt{6}}{2}$ | $x - 1 = \frac{\sqrt{6}}{2}$ and $x - 1 = -\frac{\sqrt{6}}{2}$<br>$\Rightarrow 2x - 2 - \sqrt{6} = 0$ and $2x - 2 + \sqrt{6} = 0$ |

**(iii)** The axes are parallel to the coordinates axes because the equation does not contain xy term. We have

$$4x^2 + y^2 - 8x + 2y + 1 = 0$$

or $4(x^2 - 2x) + (y^2 + 2y) + 1 = 0$

or $4(x^2 - 2x + 1) + (y^2 + 2y + 1) - 5 + 1 = 0$

or $4(x - 1)^2 + (y + 1)^2 = 4$

or $$\frac{(x-1)^2}{1} + \frac{(y+1)^2}{4} = 1 \qquad \ldots(1)$$

Shifting the origin to the point (1,-1), the equation (1) will reduce to

$$\frac{X^2}{1} + \frac{Y^2}{4} = 1 \qquad \ldots(2)$$

Comparing with the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, we have

$a^2 = 1$ and $b^2 = 4 \Rightarrow a = 1, b = 2$

$\because$ b > a, so, the major axis is parallel to the Y-axis

Length of major axis = 2b = 4 unit

Length of minor axis = 2a = 2 unit

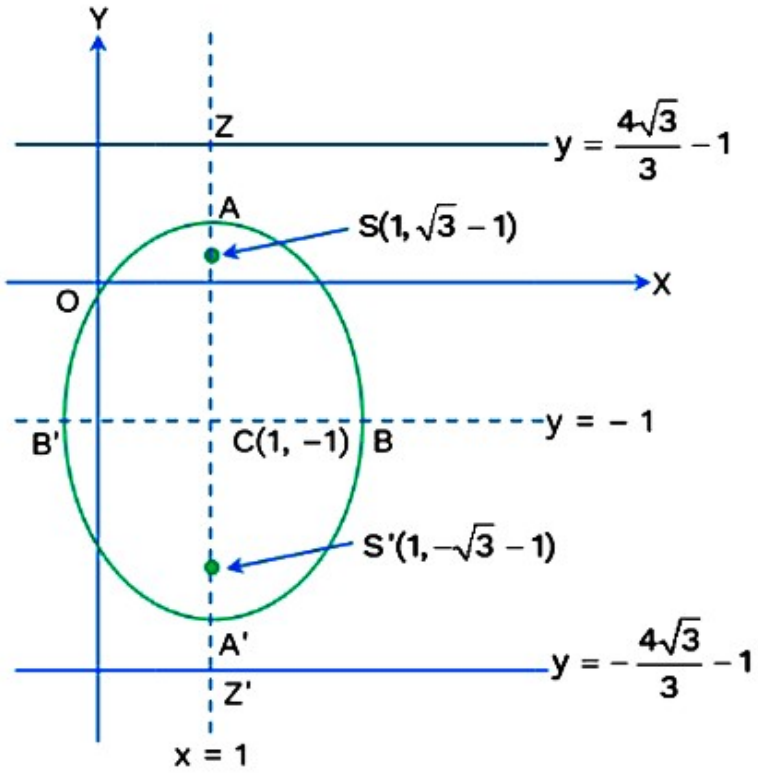
Length of latus rectum of $\frac{2a^2}{b} = \frac{2 \cdot 1}{2} = 1$ unit

Eccentricity, $e = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$

Also, $be = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$ and $\frac{b}{e} = -\frac{2}{\sqrt{3}/2} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$

Now. the coordinates of the centre and foci and the equations the axes and directrices are given in the following table:

| | X, Y System | x, y System |
|---|---|---|
| Coordinates of centre | (0, 0), i.e, X = 0, Y = 0 | $x - 1 = 0$ and $y + 1 = 0$ $\Rightarrow x = 1, y = -1$ $\therefore$ Coordinates of centre are $(1, -1)$ |
| Coordinates of foci | (0, be) and (0, – be) i.e. $(0, \sqrt{3})$ and $(0, -\sqrt{3})$ | $x - 1 = 0, y + 1 = \sqrt{3}$ and $x - 1 = 0 \quad y + 1 = -\sqrt{3}$ $\Rightarrow x = 1, y = -1 + \sqrt{3}$ and $x = 1, y = -1 - \sqrt{3}$ $\therefore$ Foci are $(1, -1 + \sqrt{3})$ and $(1, -1 - \sqrt{3})$ |
| Equation of major axis | $X = 0$ | $x - 1 = 0$ |
| Equation of minor axis | $Y = 0$ | $y + 1 = 0$ |
| Equation of directrices | $Y = \frac{b}{e}$ and $Y = -\frac{b}{e}$ i.e. , $Y = \frac{4\sqrt{3}}{3}$ and $Y = -\frac{4\sqrt{3}}{3}$ | $y + 1 = \frac{4\sqrt{3}}{3}$ and $y + 1 = -\frac{4\sqrt{3}}{3}$ $\Rightarrow \quad 3y + 3 - 4\sqrt{3} = 0$ and $3y + 3 + 4\sqrt{3} = 0$ |




**Q.** **(a) Find the equation of axes, directrix, co-ordinate of foci, centre, vertices, length of latus-rectum and eccentricity of an ellipse $\frac{(x-3)^2}{25} + \frac{(y-2)^2}{16} = 1$.**

**(b) Find the centre, the length of the axes, and the eccentricity of the ellipse $2x^2 + 3y^2 - 4x + 12y + 13 = 0$**

**(c) Find the centre, the length of the axes, eccentricity and the foci of the ellipse. $12x^2 + 4y^2 + 24x - 16y + 25 = 0$**

**Sol.** **(a)** Let x −3 = X, y −2 = Y, so equation of ellipse becomes as $\frac{X^2}{5^2} + \frac{Y^2}{4^2} = 1$

a = 5, b = 4 and a > b

equation of major axis is Y = 0 ⇒ y=2.

equation of minor axis is X = 0 ⇒ x=3.

centre (X = 0, Y = 0) & ⇒ x =3,y = 2 C ≡ (3, 2)

Length of major axis 2a = 10 Length of minor axis = 2b = 8.

Let 'e' be eccentricity, then

$\therefore\ b^2 = a^2\left(1 - e^2\right)$

$$e = \sqrt{\frac{a^2 - b^2}{a^2}} = \sqrt{\frac{25 - 16}{25}} = \frac{3}{5}$$

Length of latus rectum $= LL' = \frac{2b^2}{a} = \frac{2 \times 16}{5} = \frac{32}{5}$

Co-ordinates foci are $X = \pm ae$, $Y = 0$

$x - 3 = \pm 5 \cdot \frac{3}{5}$ and $y - 2 = 0 \Rightarrow x = 3 \pm 3$ and $y = 2$ i.e. (6, 2) and (0, 2)

**(b)** $2x^2 + 3y^2 - 4x + 12y + 13 = 0$

$2(x^2 - 2x) + 3(y^2 + 4y) + 13 = 0$

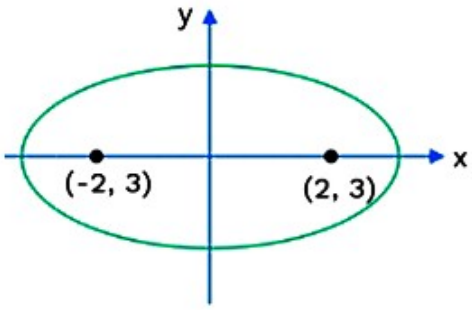
$2(x-1)^2 + 3(y + 2)^2 = 1$

$$\frac{(x - 1)^2}{\left(\frac{1}{2}\right)} + \frac{(y + 2)^2}{\left(\frac{1}{3}\right)} = 1$$

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1 \ ;$$

where

$X = x - 1$, $Y = y + 2$, $a = \frac{1}{\sqrt{2}}$, $b = \frac{1}{\sqrt{3}}$ & $a > b$




Centre is: $X = 0$, $Y = 0$; $x = 1$, $y = -2$

Length of major axis $= 2a = \sqrt{2}$ and Length of minor axis $= 2b = \frac{2}{\sqrt{3}}$

If e denotes the eccentricity, then, $b^2 = a^2\left(1 - e^2\right)$

$\therefore \ \frac{1}{3} = \frac{1}{2}\left(1 - e^2\right); e = \frac{1}{\sqrt{3}}$ Ans.

**(c)** The given equation can be written in the form $12(x + 1)^2 + 4(y - 2)^2 = 3$

$\Rightarrow \frac{(x + 1)^2}{1/4} + \frac{(y - 2)^2}{3/4} = 1$ here $a^2 = \frac{1}{4}$ and $b^2 = \frac{3}{4} \Rightarrow a < b$

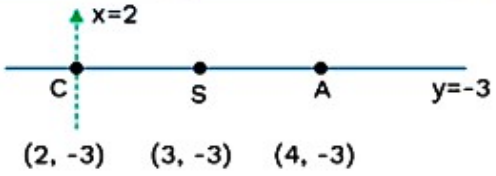
Co- ordinates of centre of the ellipse are given by $x + 1 = 0$ and $y - 2 = 0$ Hence centre of the ellipse is $(-1, 2)$ $\therefore$ Length of major axis $= 2a = \sqrt{3}$ and Length of minor axis $= 2b = 1$ i.e. $a = \frac{\sqrt{3}}{2}, b = \frac{1}{2}$ Since

$b^2 = a^2\left(1-e^2\right) \therefore \frac{1}{4} = \frac{3}{4\left(1-e^2\right)} \Rightarrow e = \sqrt{\frac{2}{3}} \therefore ae = \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{\sqrt{3}} = \frac{1}{\sqrt{2}}$

Co-ordinates of foci are given by $x + 1 = 0$, $y - 2 = \pm ae$ Thus foci are $\left(-1, 2 \pm \frac{1}{\sqrt{2}}\right)$

**Q.** **(a) Find the equation of the ellipse having its centre at the point (2, −3), one focus at (3, −3) and one vertex at (4, −3).**
**(b) Find the equation of the ellipse the extremities of whose minor axis are (3, 1) and (3, 5) and whose eccentricity is 1/2.**
**(c) Find the equation of the ellipse with its centre at (1, 2), one focus at (6, 2) and passing through the point (4, 6)**
**(d) Find the equation to the ellipse whose focus is the point (−1, 1), whose directrix is the straight line x−y + 3 = 0 and eccentricity is $\frac{1}{2}$ .**

**Sol.** **(a)** Let C = (2, −3), S = (3, −3) and A = (4, −3)




Now $CA = \sqrt{(4-2)^2 + (-3+3)^2} = 2 \therefore a = 2$

Again, $CS = \sqrt{(3-2)^2 + (-3+3)^2} = 1$

$\therefore \quad ae = 1; \qquad \therefore \qquad e = \frac{1}{a} = \frac{1}{2} \Rightarrow b^2 = a^2\left(1-e^2\right) = 4\left(1-\frac{1}{4}\right) = 3$

Major axis is $y = -3$ and parallel to x-axis

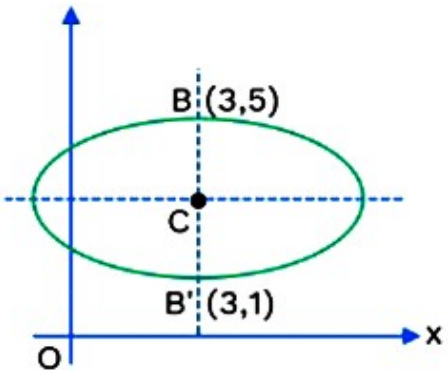
$\therefore$ Equation of ellipse is $\frac{(x-2)^2}{4} + \frac{(y+3)^2}{3} = 1$

**(b)** Let C be the centre of the ellipse. Let B' = (3,1) and B = (3,5), then C = (3, 3)
[Since C is the mid-point of BB']
Also $BB' = 4 \therefore 2b = 4 \Rightarrow b = 2$

Also, slope of $BB' = \frac{5-1}{0}$ (not defined)

Hence minor axis is parallel to y-axis and therefore, major axis will be parallel to x -axis. Let a be the length of semi-major axis of the ellipse, then $b^2 = a^2(1-e^2)$

$\therefore 4 = a^2\left(1-\frac{1}{4}\right)$ or $a^2 = \frac{16}{3}$



Since centre of the ellipse is (3, 3), therefore, its equation will be

$$\frac{(x-3)^2}{16/3}+\frac{(y-3)^2}{4}=1$$

or $3x^2+4y^2-18x-24y+47=0$

**(c)** Let S ≡ (6, 2) and C ≡ (1, 2). Slope of CS = 0, therefore major axis of the ellipse is parallel to x-axis and minor axis is parallel to y -axis. Since centre of the ellipse is (1, 2), therefore its equation will be of the form $\frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{b^2}=1$

Since (4,6) lies on (1), therefore $\frac{(4-1)^2}{a^2}+\frac{(6-2)^2}{b^2}=1$ or $\frac{9}{a^2}+\frac{16}{b^2}=1$

Since ae = distance between centre and focus = 5,

$(\therefore b^2=a^2(1-e^2)=a^2-(ae)^2=a^2-25 \quad \ldots (3)$

Substituting this value in (2), we have

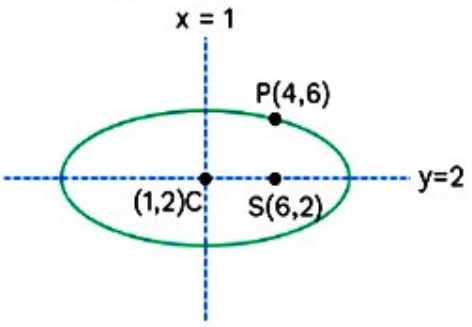
$\frac{9}{a^2}+\frac{16}{a^2-25}=1$ "or " $a^4-50a^2+225=0$

or $(a^2-45)(a^2-5)=0$ or $a^2=45, 5$.

When $a^2=5$, from (3),$b^2<0$ (not possible)

$\therefore a^2=45$, and from (3),$b^2=45-25=20$

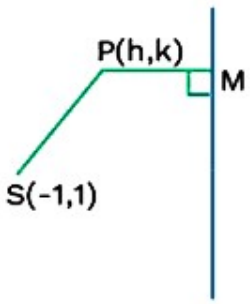
Hence from (1), equation of required ellipse is $\frac{(x-1)^2}{45}+\frac{(y-2)^2}{20}=1$




**(d)** Let P ≡ (h , k) be moving point,

$e=\frac{PS}{PM}=\frac{1}{2}$

$\Rightarrow \quad (h+1)^2+(k-1)^2=\frac{1}{4}\left(\frac{h-k+3}{\sqrt{2}}\right)^2$




$\Rightarrow$ locus of P (h, k) is

$8\{x^2+y^2+2x-2y+2\}=(x^2+y^2-2xy+6x-6y+9)$

$7x^2+7y^2+2xy+10x-10y+7=0.$

**Q.** **The positive difference between the lengths of the latus rectum of $3y=x^2+4x-9$ and $x^2+4y^2-6x+16y=24$ is**

**(A)** $\frac{1}{2}$ **(B) 2** **(C)** $\frac{3}{2}$ **(D)** $\frac{5}{3}$

**Sol.** **(A)**

$$(x+2)^2 = 3\left(y+\frac{13}{3}\right)$$

Length of latus rectum of parabola
$L_1 = 3$

$$\frac{(x-3)^2}{49}+\frac{(y+2)^2}{(49/4)}=1$$

Lengths of latus rectum of ellipse,

$$L_2 = \frac{2b^2}{a} = \frac{2\left(\frac{49}{4}\right)}{7} = 7/2$$

$$|L_1 - L_2| = \left|3-\frac{7}{2}\right| = 1/2$$

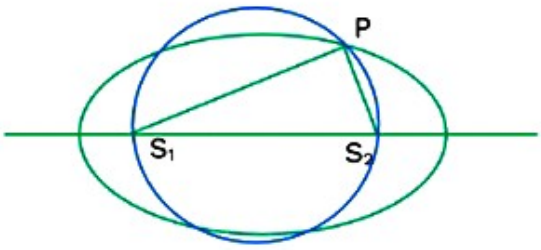
**Q.** **An circle concentric to the ellipse $\frac{4x^2}{289}+\frac{y^2}{\lambda^2}=1\left(\lambda<\frac{17}{2}\right)$ passes through foci $S_1$, $S_2$ and cuts ellipse at point P. If area of $\Delta PS_1S_2$ is 30 sq. units then find $\frac{x}{3}-\frac{y}{5}=1$**

**Sol.** area of $\Delta PS_1S_2 = \frac{1}{2}(PS_1)(PS_2) = 30$

$\Rightarrow (PS_1)(PS_2) = 60$
$PS_1 = S_2P = 2a = 17$

$$(S_1S_2) = PS_1^2 + PS_2^2 = (17)^2 - 2(60) = 169$$

$$\therefore \frac{S_1S_2}{13} = 1$$




**Q.** **Find the eccentricity of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose latus rectum is half of its major axis.**

**Sol.** According to question $\frac{2b^2}{a} = a$

$\Rightarrow \quad 2b^2 = a^2 \Rightarrow 2a^2(1-e^2) = a^2 \Rightarrow \quad e = \frac{1}{\sqrt{2}}$

**Q.** **If the focal distance of the end of minor axis of an ellipse is 2q and distance between its foci is 2p, then find its equation.**

**Sol.** Let the equation of ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

According to question $2ae = 2p$ ...(i)

and $a = q$

$\Rightarrow b^2 = a^2 - a^2e^2 \Rightarrow b^2 = q^2 - p^2$

$\Rightarrow$ Equation is $\frac{x^2}{q^2}+\frac{y^2}{q^2-p^2}=1$ Ans.

**Q.** **(a) Find the equation of the ellipse having axes along the coordinate axes and passing through the points (4, 3) and (–1, 4).**
**(b) Find the equation to the ellipse whose centre is origin, axes are the axes of co-ordinate and passes through the points (2, 2) and (3, 1).**

**Sol.** **(a)** Let the equation of the required ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ...(1)

(4, –3) is on the ellipse $\therefore \frac{4^2}{a^2}+\frac{(-3)^2}{b^2}=1$ or $\frac{16}{a^2}+\frac{9}{b^2}=1$ ...(2)

(–1, 4) is also on the ellipse $\therefore \frac{(-1)^2}{a^2}+\frac{4^2}{b^2}=1$ or $\frac{1}{a^2}+\frac{16}{b^2}=1$ ...(3)

$(3)\times 16 \Rightarrow \frac{9}{a^2}+\frac{256}{b^2}=16$ ...(4)

$(4)-(2) \Rightarrow \frac{247}{b^2}=15 \Rightarrow b^2=\frac{247}{15}$

From (3), $\frac{1}{a^2}+16\left(\frac{15}{247}\right)=1 \Rightarrow \frac{1}{a^2}=1-\frac{240}{247}=\frac{7}{247} \Rightarrow a^2=\frac{247}{7}$

$\therefore$ The equation of the required ellipse is

$\frac{x^2}{\frac{247}{7}}+\frac{y^2}{\frac{247}{15}}=1$ or $7x^2+15y^2=247$

**(b)** Let the equation to the ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ Since it passes through the points (2, 2) and (3, 1)

$\therefore \frac{4}{a^2}+\frac{4}{b^2}=1$ ...(i)

and $\frac{9}{a^2}+\frac{1}{b^2}=1$ ...(ii)

from (i) and (ii), we get

$a^2=\frac{32}{3}$ and $b^2=\frac{32}{5}$

**Q.** **(a) Find the equation of curve whose parametric equation are $x = 1 + 4\cos\theta$, $y = 2 + 3\sin\theta$. Also find its eccentricity.**

**(b) Find the eccentric angle of the point on the ellipse $\frac{x^2}{6}+\frac{y^2}{2}=1$ whose distance from the centre is 2.**

**(c) Find the eccentric angles of the extremity of latus rectum lie in the first quadrant of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.**

**Sol.** **(a)** $\cos\theta=\frac{x-1}{4}$ and $\sin\theta=\frac{y-2}{3}$

$\because \sin^2\theta+\cos^2\theta=1 \Rightarrow \frac{(x-1)^2}{16}+\frac{(y-2)^2}{9}=1$

$\therefore e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{9}{16}}=\frac{\sqrt{7}}{4}$ Ans.

**(b)** Any point on the ellipse is $(\sqrt{6}\cos\phi,\sqrt{2}\sin\phi)$, where $\phi$ is an eccentric angle of the point. It's distance from the centre (0, 0) is given and equal to 2, therefore

$6\cos^2\phi+2\sin^2\phi=4$ or $3\cos^2\phi+\sin^2\phi=2$

$2\cos^2\phi=1 \Rightarrow \cos\phi=\pm\frac{1}{\sqrt{2}}$ ; $\phi=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$ Ans.

**(c)** The coordinate of any point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose eccentric angle $\theta$ are (a cos θ, b sin θ). The coordinate of the end point of latus rectum are $\left(ae,\pm\frac{b^2}{a}\right)$

$\therefore$ For 1st quadrant

$a\cos\theta=ae$ and $b\sin\theta=\frac{b^2}{a}$ ; $\tan\theta=\frac{b}{ae} \Rightarrow \theta=\tan^{-1}\left(\frac{b}{ae}\right)$.

**Q.** **The eccentricity of the ellipse which meets the straight line $\frac{x}{7}+\frac{y}{2}=1$ on the axis of x and the straight line $\frac{x}{3}-\frac{y}{5}=1$ on the axis of y and whose axis lie along the axes of coordinates is**

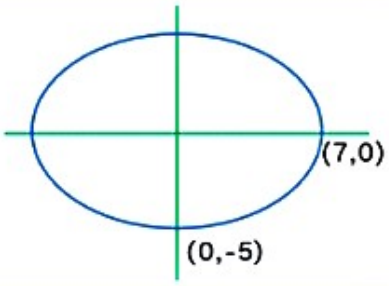
**(A)** $\frac{3\sqrt{2}}{7}$ **(B)** $\frac{2\sqrt{6}}{7}$ **(C)** $\frac{2\sqrt{3}}{7}$ **(D) None of these**

**Sol.** **(B)**

$b^2 = a^2(1 - e^2)$

$25 = 49(1 - e^2)$

$e = \frac{2\sqrt{6}}{7}$




**Q.** **(a) A rod of length 12cm moves with its ends always touching the co-ordinate axes. Determine the equation of the locus of a point on the rod, which is 3cm from the end in contact with the x-axis.**

**(b) A line of fixed length a + b moves so that its ends are always on two fixed perpendicular lines; prove that the locus of a point, which divide this line into two portions of lengths 'a' and 'b' is an ellipse.**

**Sol.** **(a)** Let AB be the rod of length 12cm touching the co-ordinate axes at points A and B.

Let A ≡ (a, 0), B ≡ (0, b)

Now $AB^2 = 12^2$

$\Rightarrow a^2 + b^2 = 144$ ...(1)

Let P be a point on AB such that

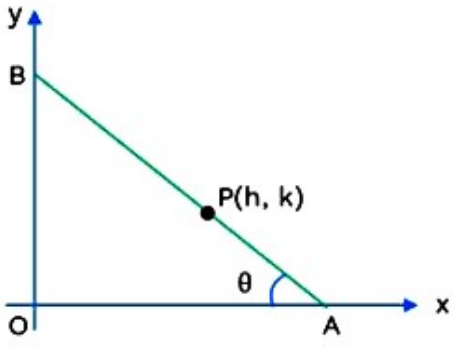
AB = 3cm

then BP = 12cm −3cm = 9cm

$\therefore$ AB : PB = 1 : 3

Hence P divides AB internally in the ratio 1 : 3

$\therefore \quad P \equiv \left(\frac{1.0+3.a}{1+3}, \frac{1.b+3.0}{1+3}\right)$ or $P \equiv \left(\frac{3a}{4}, \frac{b}{4}\right)$




$\therefore \quad h = \frac{3a}{4}$ and $k = \frac{b}{4} \Rightarrow a = \frac{4h}{3}$ and $b = 4k$

Put value of a and b in equation (1) we get $\frac{16h^2}{9} + 16k^2 = 144$

$\therefore$ Locus of P(h, k ) is $\frac{16}{9}x^2 + 16y^2 = 144$,

i.e. $x^2 + 9y^2 = 81$, which is the equation of the required locus.

**(b)** Let us assume that the two fixed perpendicular lines are X-axis and Y-axis. Let the line be AB with the end B on the X-axis and the end A on the Y-axis and P be the point (h, k) on the line so that A P = a and PB = b.

If AB makes an angle $\phi$ with the X-axis as shown in the figure, then

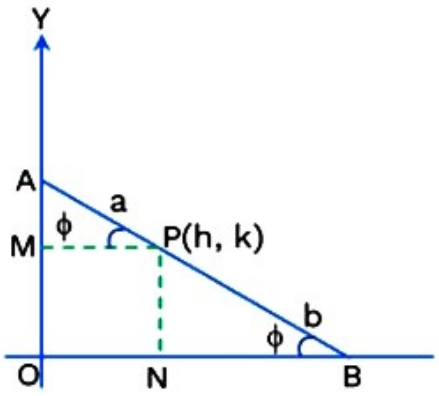
In the triangle APM.

MP = AP cos $\phi$ or h = a cos $\phi$ ...(1)

In the triangle PBN,

PN = PB sin $\phi$ or k = b sin $\phi$ ...(2)

Eliminating $\phi$ from the equations (1) and (2), we ge




$$\left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 = 1$$

Hence, the locus of P(h,k) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Which is an ellipse.

**Alternatively:**

The coordinates of A and B are, say (0, $\alpha$) and ($\beta$, 0), then

$\alpha^2 + \beta^2 = (a + b)^2$ ...(3)

Now, $\frac{AP}{PB} = \frac{a}{b}$

$\therefore$ $h = \frac{a\beta}{a+b}$ and $k = \frac{b\alpha}{a+b}$

[$\because$ P divides AB in ratio a : b internally]

$\therefore$ $\left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 = \frac{\alpha^2 + \beta^2}{(a+b)^2} = 1$ [From the equation (2)]

Thus, the required locus of P(h, k) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

**Q.** **A series of ellipse $E_1, E_2, \ldots E_n$ are drawn such that $E_n$ touches $E_{n-1}$ at the extremities at the extremities of the major axis of $E_{n-1}$ and the foci of $E_n$ coincided with the extremities of minor axis of $E_{n-1}$. If eccentricity of the ellipse is independent of n, then the value of eccentricity is**

**(A)** $\frac{\sqrt{5}}{3}$ **(B)** $\frac{\sqrt{5}+1}{4}$ **(C)** $\frac{\sqrt{5}-1}{2}$ **(D)** $\frac{\sqrt{5}-1}{3}$

**Sol.** **(C)**

$b' = a,\ a'e = b$

$b'^2 = a'^2 - a'^2e^2$

$$a^2 = \frac{b^2}{e^2} - b^2$$

$$e^2 = \frac{b^2}{a^2}\left(1-e^2\right) = \left(1-e^2\right)^2$$

$1 - e^2 = \pm e$

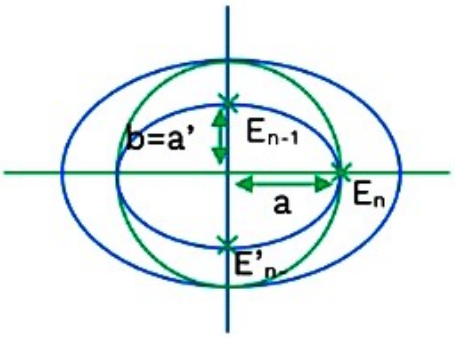
$e^2 + e - 1 = 0$

or $e^2 - e - 1 = 0$

$$\Rightarrow e = \frac{-1 \pm \sqrt{5}}{2}$$

$$\text{or } \frac{1 \pm \sqrt{5}}{2}$$

$$\Rightarrow e = \frac{\sqrt{5}-1}{2}$$




**Q.** **Two circles are given such that one is completely lying inside other without touching. Prove that the locus of the center of variable circle which touches smaller circle from outside and bigger circle from inside is an ellipse.**

**Sol.** In the figure, circles with hard line are given circles with centers $C_1$ and $C_2$ and radius $r_1$ and $r_2$ respectively. Let the circle with dotted line is a variable circle which touches given two circles as explained in the question which has center C and radius r.

Now $CC_2 = r + r_2$ and $CC_1 = r_1 - r$

Hence, $CC_1 + CC_2 = r_1 + r_2 = \text{(constant)}$

Then locus of C is ellipse whose foci are $C_1$ and $C_2$.

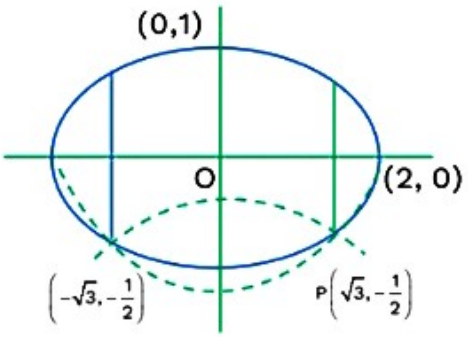
**Q.** **Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$ be the end points the latus rectum of ellipse $x^2 + 4y^2 = 4$. The equations of parabolas with latus rectum PQ are :**

**(A)** $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$ **(B)** $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$

**(C)** $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$ **(D)** $x^2 - 2\sqrt{3}y = 3 - \sqrt{3}$

**Sol.** **(BC)**

$1 = 4\left(1 - e^2\right) \Rightarrow e = \frac{\sqrt{3}}{2}$




Length of latus rectum of parabola

$= 4a = 2\sqrt{3}$

$\Rightarrow a = \frac{\sqrt{3}}{2}$

Equation of parabolas are

$2\sqrt{3}\left(y - \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\right)\right) = x^2$

or $-2\sqrt{3}\left(y - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\right)\right) = x^2$

$2\sqrt{3}y + (1 + \sqrt{3})\sqrt{3} = x^2$

or $-2\sqrt{3}y - (1 - \sqrt{3})\sqrt{3} = x^2$

or $x^2 - 2\sqrt{3}y = \sqrt{3} + 3$

$x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$

**Q.** **If (5, 12) and (24, 7) are the foci of an ellipse passing through origin, then find the eccentricity of the ellipse.**

**Sol.** Let the S(5,12) & S(24,7) are two foci & ellipse passes through origin O.

$\therefore$ OS + OS' = 2a

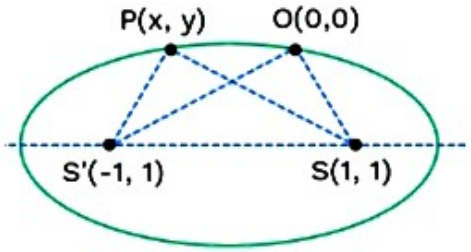
$\Rightarrow \sqrt{25 + 144} + \sqrt{576 + 49} = 2a$

$\Rightarrow$ 2a = 13 + 25 $\Rightarrow$ a = 19 and $2ae = SS' = \sqrt{386}$

$\therefore e = \frac{\sqrt{386}}{38}$

**Q.** **Find the equation of the ellipse that passes through the origin and has the foci at the points (-1,1) and (1,1).**

**Sol.** Let P(x,y) be any point on the ellipse and the foci be S(−1,1) and S' (1, 1). and O lie on the ellipse




OS + OS' = constant = 2a

$\Rightarrow \sqrt{(x+1)^2+(y-1)^2}+\sqrt{(x-1)^2+(y-1)^2}=2\sqrt{2}$

Let P(x, y) be any point on the ellipse

$\Rightarrow$ PS + PS' = $2\sqrt{2}$

$\Rightarrow \sqrt{(x+1)^2+(y-1)^2}+\sqrt{(x-1)^2+(y-1)^2}=2\sqrt{2}$

$\Rightarrow \sqrt{(x+1)^2+(y-1)^2}=2\sqrt{2}-\sqrt{(x-1)^2+(y-1)^2}$

$\Rightarrow (x+1)^2+(y-1)^2=8+\left[(x-1)^2+(y-1)^2\right]-4\sqrt{2}\sqrt{(x-1)^2+(y-1)^2}$

$\Rightarrow x^2+2x+1+y^2-2y+1=8+x^2-2x+1+y^2-2y+1-4\sqrt{2}\sqrt{(x-1)^2+(y-1)^2}$

$\Rightarrow 4x-8=-4\sqrt{2}\sqrt{(x-1)^2+(y-1)^2}$

$\Rightarrow x-2=-\sqrt{2}\sqrt{(x-1)^2+(y-1)^2}$

$\Rightarrow x^2-4x+4=2\left[x^2-2x+1+y^2-2y+1\right]$

$\Rightarrow 0=x^2+2y^2-4y$

$\Rightarrow x^2+2y^2-4y=0$ is the required equation of the ellipse.

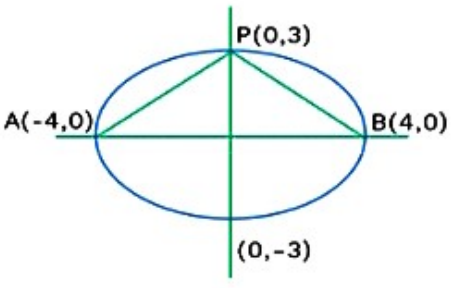
**Q.** **If base of a triangle is the major axis of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and third vertex moves on the ellipse, then maximum area of triangle will be :**

**(A) 6** **(B) 72** **(C) 12** **(D) None of these**

**Sol.** **(C)**

$\Rightarrow$ P(0, 3) or P(0, − 3)

Area of $\Delta$PAB is maximum when altitude is maximum. Hence P should be (0, 3)

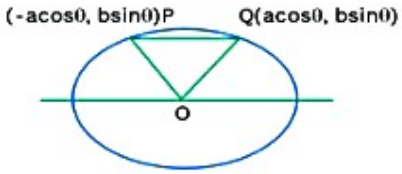



$(\Delta PAB)_{max}=\frac{1}{2}\times 8\times 3=12$

**Q.** **P and Q are two points on the upper half of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. The centre of the ellipse is at the origin 'O' and PQ is parallel to the x–axis such that the triangle OPQ has the maximum possible area. A point is randomly selected from inside of the upper half of the ellipse. The probability that is lies outside the triangle, is :**

**(A)** $\frac{\pi-1}{\pi}$ **(B)** $\frac{2\pi-1}{2\pi}$ **(C)** $\frac{\pi-1}{2\pi}$ **(D)** $\frac{\pi-1}{4\pi}$

**Sol.** **(A)**




Area of $\Delta OPQ = \frac{1}{2}(2a\cos\theta)(b\sin\theta)$

$\frac{ab}{2}\sin 2\theta \le \frac{ab}{2}$

$$\text{Probability} = \frac{\frac{\pi ab}{2}-\frac{ab}{2}}{\frac{\pi ab}{2}} = \frac{\pi-1}{\pi}$$

**Q.** **Find the equation of the ellipse having major and minor axes along x and y axes respectively, the distance between whose foci is 8 units and the distance between the directrices is 18 units.**

**Sol.** Given, 8 = distance between foci = 2ae ...(1)

and 18 = distance between directrices $= \frac{2\alpha}{e}$ ...(2)

(1) × (2) ⇒ (8) (18) = (2ae)$\left(\frac{2a}{e}\right)$ = $4a^2$ ⇒ $a^2$ = 36 ⇒ a = 6 ...(3)

Again, 8 = 2ae = 2(6) e = 12e

$\Rightarrow e = \frac{8}{12} = \frac{2}{3}$

Also,

$$b^2 = a^2\left(1-e^2\right) = (6)^2\left[1-\left(\frac{2}{3}\right)^2\right] = 36\left(\frac{5}{9}\right) = 20$$

∴ Required equation of the ellipse is

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ or $\frac{x^2}{36}+\frac{y^2}{20}=1$ [From (3) and (5)

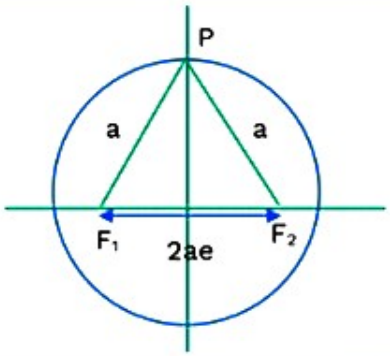
**Q.** **If the line joining foci subtends an angle of 90° at an extremity of minor axis then the eccentricity of the ellipse is :**

**(A)** $\frac{1}{\sqrt{3}}$ **(B)** $\frac{1}{\sqrt{2}}$ **(C)** $\frac{1}{2}$ **(D) None of these**

**Sol.** **(B)**

$a^2 + a^2 = (2ae)^2$

$\Rightarrow e^2 = \frac{1}{2} \Rightarrow e = \frac{1}{\sqrt{2}}$




**Q.** **The locus of incentre of $\Delta PF_1F_2$ where P is a variable point lying on ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with $F_1$, $F_2$ as foci and eccentricity e is a conic whose:**

**(A) equation is** $\frac{x^2}{a^2e^2}+\frac{y^2(1+e)^2}{b^2e^2}=1$ **(B) equation is** $\frac{x^2}{a^2e^2}+\frac{y^2(1-e)^2}{b^2e^2}=1$

**(C) eccentricity is** $\sqrt{\frac{2e}{1+e}}$ **(D) eccentricity is** $\sqrt{\frac{2e}{1-e}}$

**Ans.** **(AC)**

$$h = \frac{a^2e(1+e\cos\theta) - a^2e(1-e\cos\theta) + 2a^2e\cos\theta}{2a + 2ae}$$

$\Rightarrow h = ae\cos\theta$

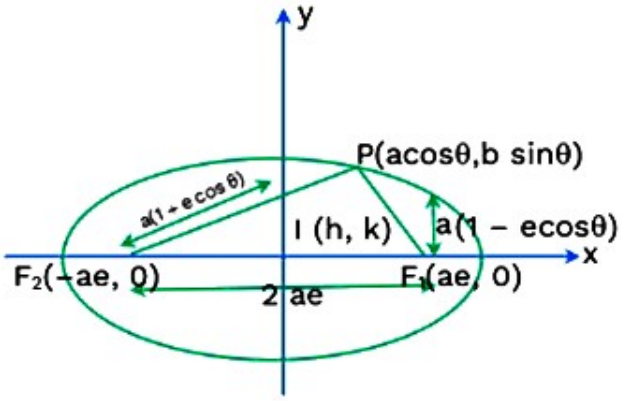
$$k = \frac{2aeb\sin\theta}{2a+2ae} = \frac{eb\sin\theta}{(1+e)}$$

$$\therefore \quad \frac{h^2}{(ae)^2} + \frac{k^2}{\left(\frac{be}{1+e}\right)^2} = 1$$

Equation of locus of incentre is

$$\frac{x^2}{(ae)^2} + \frac{y^2}{\left(\frac{be}{1+e}\right)^2} = 1$$

$$\therefore \quad \frac{b^2e^2}{(1+e)^2} = a^2e^2\left(1-e^2\right)^2$$



$$\Rightarrow \frac{a^2e^2\left(a-e^2\right)}{(1+e)^2} = a^2e^2\left(1-e'\right)^2$$

$$\Rightarrow\ e'^2 = 1-\frac{1-e}{1+e} = \frac{2e}{1+e} \Rightarrow\ e' = \sqrt{\frac{2e}{1+e}}$$

**Q.** **If the equation of the curve on reflection of the ellipse $\frac{(x-4)^2}{16}+\frac{(y-3)^2}{9}=1$ about the line x–y–2=0 is $16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$, then $\frac{k_1+k_2}{22}$ is?**

**Sol.** **(6)**

Image of (h, k) on ellipse about x – y – 2 = 0 is say (h', k')

$$\therefore\ \frac{h'-h}{1} = \frac{k'-k}{-1}$$

$$= -2\frac{h-k-2}{1+1} = -h+k+2$$

$\Rightarrow$ h' = k + 2, k' = h – 2

(h, k) $\equiv$ (k' + 2, h' – 2)

$$\because\ \frac{(h-4)^2}{16}+\frac{(k-3)^2}{9}=1$$

$$\Rightarrow \frac{(k'+2-4)^2}{16}+\frac{(h'-2-3)^2}{9}=1$$

$\Rightarrow 16(h')^2 + 9k'^2 - 36k' - 160h' + 292 = 0$

$\Rightarrow$ Equation of relfection of ellipse is

$$\frac{k_1+k_2}{22} = \frac{292-160}{22} = 6$$

**Q.** **In a triangle ABC with fixed base BC, the vertex A moves that cosB + cosC $=4\sin^2 A/2$. If a, b and c denote the lengths of the sides of the triangle opposite to the angles A, B and C, respectively, then**

**(A) b + c = 4a**

**(B) b +c = 2a**

**(C) locus of point A is an ellipse**

**(D) locus of point A is a pair of straight line**

**Sol.** **(BC)**

$$\cos B + \cos C = 4\sin^2\frac{A}{2}$$

$$2\cos\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right) = 4\sin^2\frac{A}{2}$$

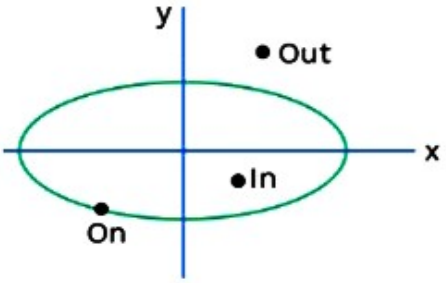
$$\Rightarrow \cos\left(\frac{B-C}{2}\right) = 2\sin\left(\frac{A}{2}\right)$$

$$\Rightarrow \frac{\cos\left(\frac{B-C}{2}\right)}{\sin\frac{A}{2}} = 2 \Rightarrow \frac{\sin B + \sin C}{\sin A} = 2$$

b + c = 2a(constant)
Hence, locus of vertex A is ellipse with B and C as foci.

**POSITION OF A POINT w.r.t. AN ELLIPSE:**

Let $S(x,y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$ be the given ellipse and $P(x_1, y_1)$ is the given point.




(i) If $S(x_1,y_1) > 0$ then $P(x_1,y_1)$ lie outside the ellipse.

(ii) If $S(x_1,y_1) < 0$ then $P(x_1,y_1)$ lie inside the ellipse.

(iii) If $S(x_1,y_1) = 0$ then $P(x_1, y_1)$ lie on the ellipse.

This result holds true for circle and parabola also.

**Q.** **Check whether the point P(3, 2) lies inside, on or outside of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$.**

**Sol.** $S(3,2) \equiv \frac{9}{25} + \frac{4}{16} - 1 = \frac{9}{25} + \frac{1}{4} - 1 < 0$

∴ Point P(3, 2) lies inside the ellipse.

**Q.** **Find the set of values of 'α' for which the point P(α,−α) lies inside the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.**

**Sol.** If P(α,−α) lies inside the ellipse then

$$\therefore\ S(\alpha,-\alpha)<0 \Rightarrow \frac{\alpha^2}{16}+\frac{\alpha^2}{9}-1<0 \Rightarrow \frac{25}{144}\cdot\alpha^2<1$$

$$\Rightarrow\quad \alpha^2<\frac{144}{25};\ \therefore \alpha\in\left(-\frac{12}{5},\frac{12}{5}\right).$$

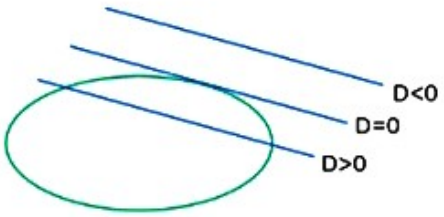
## INTERACTION OF A LINE AND AN ELLIPSE:

Let the equations of the line is

$$y = mx + c \qquad \ldots (1)$$

and equation of ellipse is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \qquad \ldots(2)$$

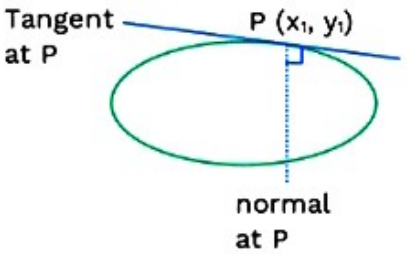



The points of intersection of the line and the ellipse can be obtained by solving the two equations simultaneously. Hence by eliminating y from (1) and (2), we get

$$\frac{x^2}{a^2}+\frac{(mx+c)^2}{b^2}=1$$

i.e. $(b^2 + a^2m^2)\,x^2 + 2a^2\,cmx + a^2\,(c^2-b^2) = 0$ ...(3)

Let $x_1$, $x_2$ be the roots of the quadratic equation (3). The line meets the ellipse in real and distinct points if the roots $x_1$ and $x_2$ are real and different. The line is a tangent to the ellipse if $x_1 = x_2$ and the line does not meet the ellipse if the roots $x_1$ and $x_2$ are imaginary. All these will be decided by the discriminant of quadratic equation (3).



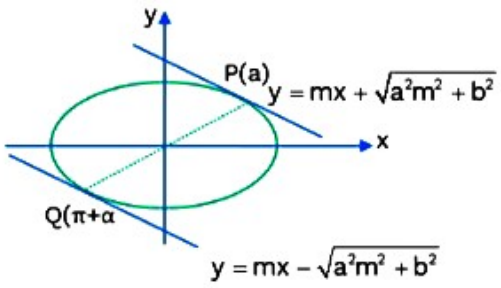

## TANGENTS :

### (i) Point form:

$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ is tangent to the ellipse at $(x_1, y_1)$ Since point $(x_1, y_1)$ lie on the curve therefore we can use standard substitution to obtain the equation of tangent.

**(ii) Slope form:**

Let the given line is $y = mx + c$ and given ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.




If line touch is the ellipse then by solving the two equations simultaneously (by eliminating y from

(1) & (2)), we get $\frac{x^2}{a^2} + \frac{(mx + c)^2}{b^2} = 1$.

i.e. $\left(b^2 + a^2m^2\right)x^2 + 2a^2cmx + a^2\left(c^2 - b^2\right) = 0$

Since line is tangent to the ellipse therefore its D = 0

or $4a^4c^2m^2 - 4(b^2 + a^2m^2).a^2(c^2 - b^2) = 0$

or $4a^2 [a^2c^2m^2 - b^2c^2 - a^2c^2m^2 + b^4 + a^2b^2m^2] = 0$

or $b^2(-c^2+n^2 + a^2m^2) = 0$

which is the required condition of tangency.

Substituting this value of c in $y = mx + c$, we have $y = mx + \sqrt{a^2m^2 + b^2}$ or

$y = mx - \sqrt{a^2m^2 + b^2}$, which are tangents to the ellipse for all values of mHere ± sign represents two tangents to the ellipse having the same m, i.e. there are two tangents parallel to any given direction.

The equation of any tangent to the ellipse

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$(y-k) = m(x-h) \pm \sqrt{a^2m^2 + b^2}$$

**(iii) Parametric form:**

$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$ is tangent to the ellipse at the point (a cos θ, b sin θ).

**Q.** **Find the equations of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.**

**Sol.** Slope of tangent to the given line $=-2$ Given ellipse is $\frac{x^2}{4}+\frac{y^2}{3}=1$

Equation of tangent whose slope is 'm' is $y = mx \pm \sqrt{4m^2+3}$

$\because m = \frac{1}{2} \therefore y = \frac{1}{2}x \pm \sqrt{1+3}$

$2y = x \pm 4$

**Q.** **A tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ touches at the point P on it in the first quadrant and meets the co-ordinate axes in A and B respectively. If P divides AB in the ratio 3 : 1, find the equation of the tangent.**

**Sol.** Let P = $(a\cos\theta, b\sin\theta)$

$\therefore$ equation of tangent is $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1]$

A = $(a\sec\theta, 0)$ B = $(0, b\,\mathrm{cosec}\theta)$ $\because$ P divide AB internally in the ratio 3:1

$\therefore a\cos\theta = \frac{a\sec\theta}{4} \Rightarrow \cos^2\theta = \frac{1}{4} \Rightarrow \cos\theta = \frac{1}{2}$ and $b\sin\theta = \frac{3b\,\mathrm{cosec}\theta}{4} \Rightarrow \sin\theta = \frac{\sqrt{3}}{2}$

$\therefore$ tangent is $\frac{x}{2a}+\frac{\sqrt{3}y}{2b}=1 \Rightarrow bx + \sqrt{3}ay = 2ab$ ans.

**Q.** **Find the equations of the tangents drawn from the point (2, 3) to the ellipse $9x^2 + 16y^2 = 144$.**

**Sol.** Let the equation of the tangent is $y = mx \pm \sqrt{16m^2+9}$

It passes through the point (2, 3)

$\Rightarrow \quad 3 = 2m + \sqrt{16m^2+9}$

$\Rightarrow \quad (3-2m)^2 = 16m^2 + 9$

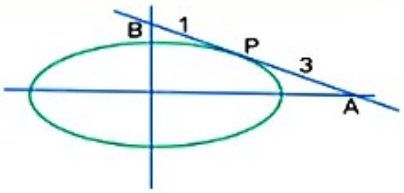
$\Rightarrow \quad 12m^2 + 12 = 0$

$\Rightarrow \quad m = 0, -1$

$\Rightarrow$ Tangent are $y = 3$ or $y = -x + 5$

**Q.** **Find the equations of the tangents to the ellipse $x^2 + 16y^2 = 16$ each one of which makes an angle of 60° with the x-axis.**

**Sol.** We have, $x^2 + 16y^2 = 16 \Rightarrow \frac{x^2}{4^2} + \frac{y^2}{1^2} = 1$




This is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$,

where $a^2 = 16$ and $b^2 = 1$ So, the equations of the tangents are

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

i.e. $y = \sqrt{3}x \pm \sqrt{16 \times 3 + 1} \Rightarrow y = \sqrt{3}x \pm 7$

**Q.** **For what value of $\lambda$ does the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$.**

**Sol.** ∵ Equation of ellipse is

$9x^2 + 16y^2 = 144$ or $\frac{x^2}{16} + \frac{y^2}{9} = 1$

Comparing this with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then we get $a^2 = 16$ and $b^2 = 9$

& comparing the line $y = x + \lambda$ with $y = mx + c$

$\therefore m = 1$ and $c = \lambda$

therefore condition of tangency $c^2 = a^2m^2 + b^2 \Rightarrow \lambda^2 = 16 \times 1^2 + 9 \Rightarrow \lambda^2 = 25$

$\therefore \lambda = \pm 5$

**Q.** **A circle of radius r is concentric with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Prove that the common tangent is inclined to the major axis at an angle $\tan^{-1}\left(\sqrt{\frac{r^2 - b^2}{a^2 - r^2}}\right)$.**

**Sol.** Let the equation of circle is $x^2 + y^2 = r^2$ ...(i)

and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ...(ii)

Tangent to be ellipse is $y = mx - \sqrt{a^2m^2 + b^2}$ i.e. $mx - y + \sqrt{a^2m^2 + b^2} = 0$. Since it is tangent to the circle also, therefore perpendicular distance from centre (0,0)

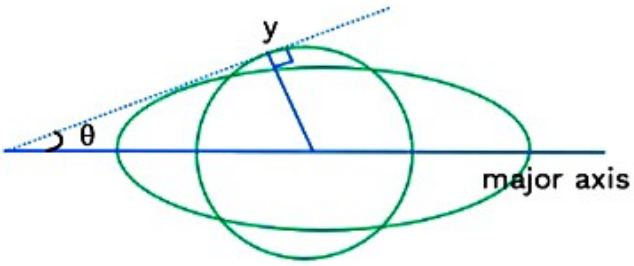
will be equal to radius r , $\therefore \left|\frac{\sqrt{a^2m^2+b^2}}{\sqrt{m^2+1}}\right| = r$

$\Rightarrow a^2m^2 + b^2 = r^2 (1 + m^2)$

$\Rightarrow m^2 = \frac{r^2-b^2}{a^2-r^2}$

$\therefore m = \tan\theta = \sqrt{\frac{r^2-b^2}{a^2-r^2}}$

$\therefore \theta = \tan^{-1}\sqrt{\frac{r^2-b^2}{a^2-r^2}}$




**Q.** **For the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ with vertices A and A', tangent drawn at the point P in the first quadrant meets the y-axis at Q and the chord A´P meets the y-axis at M. If O is the origin, then $OQ^2 - MQ^2$ equals to**

**(A) 9** **(B) 13** **(C) 4** **(D) 5**

**Sol.** **(C)**

Let point P be (acos θ, bsinθ)

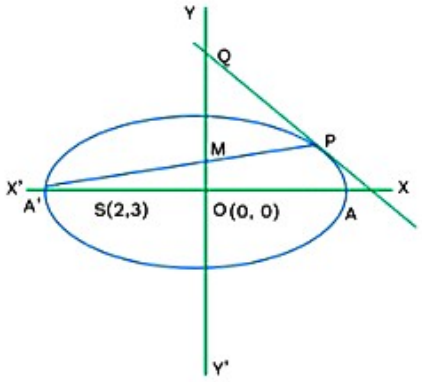
Equation of the tangent at point P is

$\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$

Then point Q is (bcosec θ, 0)

Equation of chord A'P is $y-0=\frac{b\sin\theta}{a\cos\theta+a}(x+a)$

Putting x = 0, we have $y=\frac{b\sin\theta}{\cos\theta+1}$




Then

$OQ^2 - MQ^2$

$=\frac{2b^2}{\cos\theta+1}-\frac{b^2\sin^2\theta}{(\cos\theta+1)^2}$

$=\frac{b^2}{\cos\theta+1}\left(\frac{2\cos\theta+2-\sin^2\theta}{(\cos\theta+1)}\right)$

$=\frac{b^2}{\cos\theta+1}\left(\frac{2\cos\theta+1+\cos^2\theta}{(\cos\theta+1)}\right)$

$= b^2 = 4$

**Q.** **Prove that in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ the locus of the middle points of the portions of tangents included between the axes is the curve $\frac{a^2}{x^2}+\frac{b^2}{y^2}=4$.**

**Sol.** Equation of any tangent to the given ellipse may be taken as

$$\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$$

Let this tangent meet the x -axis in P and y -axis in Q.
Putting y = 0 In (1), we get x = a secθ.
∴ Co-ordinates of P are (a secθ,0)
Similarly co-ordinates of Q are (0,b cosecθ) Let (h, k) be the mid. point of PQ.

$$\therefore\ h=\frac{a\sec\theta+0}{2}=\frac{a\sec\theta}{2} \text{ and } k=\frac{0+b\operatorname{cosec}\theta}{2}=\frac{b\operatorname{cosec}\theta}{2}$$

Hence $2\cos\theta=\frac{a}{h}$

and $2\sin\theta=\frac{b}{k}$

Squaring and adding (2) and (3), we have $4=\frac{a^2}{h^2}+\frac{b^2}{k^2}$.

∴ Locus of (h, k) is $\frac{a^2}{x^2}+\frac{b^2}{y^2}=4$.

**Comprehension for Q.1 to Q.3**

The tangent at any point P of the circle $x^2 + y^2 = 16$ meets the tangent at a fixed point A at T, and T is joined to B, the other end of the diameter through A.

**Q.1** **The locus of the intersection of AP and BT is conic whose eccentricity is**

**(A)** $\frac{1}{2}$ **(B)** $\frac{1}{\sqrt{2}}$ **(C)** $\frac{1}{3}$ **(D)** $\frac{1}{\sqrt{3}}$

**Ans.** **(B)**

**Q.2** **Sum of focal distances of any point on the curve is**

**(A) 12** **(B) 16** **(C) 20** **(D) 8**

**Ans.** **(D)**

**Q.3** **Which of the following does not change by changing the radius of the circle?**

**(A) coordinates of focii** **(B) length of major axis**

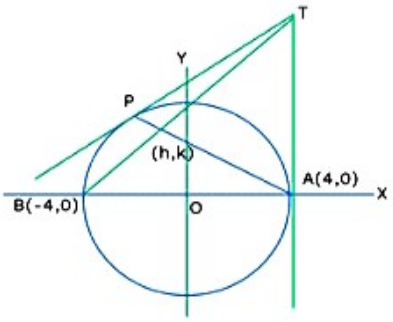
**(C) eccentricity** **(D) length of minor axis**

**Sol.** **(C)**

Tangents at P $(4\cos\theta, 4\sin\theta)$ to $x^2 + y^2 = 16$ is $x\cos\theta + y\sin\theta = 4$

Equation of AP is $y = \frac{\sin\theta}{\cos\theta - 1}(x - 4)$

From (i), coordinates of the point T are given by $\left(4, \frac{4(1-\cos\theta)}{\sin\theta}\right)$




Equation of BT is $y = \frac{1-\cos\theta}{2\sin\theta}(x + 4)$ ... (iii)

let (h, k) be the point of intersection of the lines (ii) and (iii). Then we have

$k^2 = -\frac{1}{2}(h^2 - 16)$

$\Rightarrow \frac{h^2}{16} + \frac{k^2}{8} = 1$

Therefore, locus of (h, k) is $\frac{x^2}{16} + \frac{y^2}{8} = 1$

Which is an ellipse with eccentrically $e = \frac{1}{\sqrt{2}}$

Sum of focal distances of any point = 2a = 8

Considering circle $x^2 + y^2 = a^2$. We find that the eccentricity of ellipse is $\frac{1}{\sqrt{2}}$ which is constant and does not change by changing the radius of the circle

**Q.** **Find the locus of the foot of the any tangent to perpendicular drawn from the center upon any the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$**

**Sol.** Let the foot of perpendicular is P(h, k)

Then the slope of OP is $\frac{k}{h}$ (O is centre)

$\Rightarrow$ Slope of the tangent is $-\frac{h}{k}$

$\Rightarrow$ Equation of the tangent is $y - k = -\frac{h}{k}(x - h)$

or $y = -\frac{h}{k}x + \frac{h^2+k^2}{k}$

This touches the ellipse, then $\left(\frac{h^2+k^2}{k}\right)^2 = a^2\left(\frac{h}{k}\right)^2 + b^2$

$\Rightarrow \quad a^2x^2 + b^2y^2 = \left(x^2+y^2\right)^2$

**Q.** **Common tangents are drawn to parabola $y^2 = 4x$ and the ellipse $3x^2 + 8y^2 = 48$ touching parabola at A and B and ellipse at C and D. If the area of quadrilateral of ABCD is $55\sqrt{N}$, then N =**

**Sol.** **(2)**

Equation of tangent at A is $ty = x + t^2$

$\because y = \frac{1}{t}x + t$

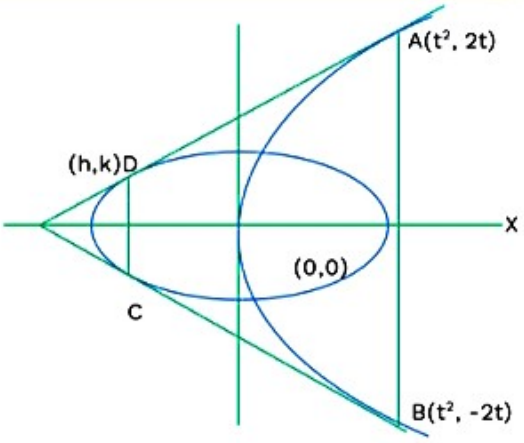
$\because$ It is tangent to ellipse also

$\Rightarrow t^2 = \frac{16}{t^2} + 6$

$\Rightarrow t^4 - 6t^2 - 16 = (t^2 - 8)(t^2 + 2) = 0$

$\Rightarrow t = \pm 2\sqrt{2}$

$\therefore A, B = (8, 4\sqrt{2})$ and $(8, -4\sqrt{2})$




Let D ≡ (h, k)

Tangent to $y^2 = 4x$ at A is $2\sqrt{2}y - x = 8$ ... (1)

Tangent to ellipse at B is $2xh + 8ky = 48$ ... (2)

$\because$ (1) and (2) are identical

$\frac{8k}{2\sqrt{2}} = -\frac{3h}{1} = \frac{48}{8} = 6$

$h = 2, \; k = \frac{3}{\sqrt{2}}$

$\Rightarrow h = -2, \quad k = \frac{3}{\sqrt{2}}$

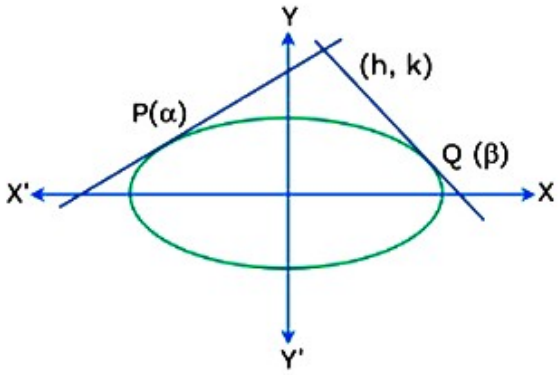
$\therefore D,.C \equiv \left(-2, \frac{3}{\sqrt{2}}\right)$ and $\left(-2, -\frac{3}{\sqrt{2}}\right)$

Area of trapezium $ABCD = \frac{1}{2}(8\sqrt{2} + 3\sqrt{2})10$

$= 55\sqrt{2} \Rightarrow N = 2$

**Q.** **Find the locus of the point of intersection of tangents to the ellipse if the difference of the eccentric angle of the points is $\frac{2\pi}{3}$.**

**Sol.** $|\alpha - \beta| = \frac{2\pi}{3}$




$$h = a\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} = 2a\cos\frac{\alpha+\beta}{2}$$

and $$k = 2b\sin\frac{\alpha+\beta}{2}$$

$$\Rightarrow \quad \frac{h^2}{4a^2} + \frac{k^2}{4b^2} = 1$$

$$\Rightarrow \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$$

**Q.** **The tangent at any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the auxiliary circle in two points which subtend a right angle at the centre. Show that the eccentricity of the ellipse is $\frac{1}{\sqrt{1+\sin^2\alpha}}$.**

**Sol.** Given ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad ...(1)$$

Its auxiliary circle is

$$x^2 + y^2 = a^2 \qquad ...(2)$$

Let $P \equiv (a\cos\alpha, b\sin\alpha)$

Equation of tangent to the ellipse at $P(a\cos\alpha, b\sin\alpha)$ is

$$\frac{x\cos\alpha}{a} + \frac{y\sin\alpha}{b} = 1 \qquad ...(3)$$

Making equation (2) homogeneous with the help of (3), we get

$$x^2 + y^2 - a^2\left(\frac{x\cos\alpha}{a} + \frac{y\sin\alpha}{b}\right)^2 = 0$$

$$\text{or } \left(1-\cos^2\alpha\right)x^2 + \left(1-\frac{a^2}{b^2}\sin^2\alpha\right)y^2 - 2\frac{a}{b}(\cos\alpha)(\sin\alpha)xy = 0 \qquad ...(4)$$

(4) is the joint equation of OL and OM.

Since $\angle LOM = 90°$

$\therefore$ Coefficient of $x^2$ + coeff. of $y^2 = 0$

$$\therefore 1-\cos^2\alpha + 1-\frac{a^2}{b^2}\sin^2\alpha = 0 \text{ or}$$

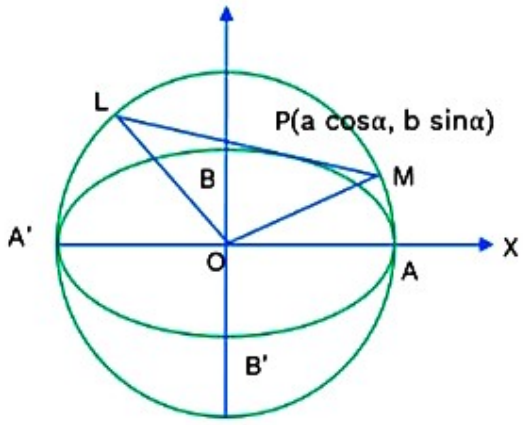
$$\sin^2\alpha\left(\frac{a^2}{b^2}-1\right) = 1$$

or

$$\sin^2\alpha\left(\frac{1}{1-e^2}-1\right) = 1\left[\because b^2 = a^2\left(1-e^2\right)\right]$$

$$\text{or } e^2\sin^2\alpha = 1-e^2 \text{ or } e^2\left(1+\sin^2\alpha\right) = 1$$

$$\text{or } e = \frac{1}{\sqrt{1+\sin^2\alpha}}$$




## NORMALS

### (i) Point form:

Equation of the tangent to the ellipse at $(x_1, y_1)$ is

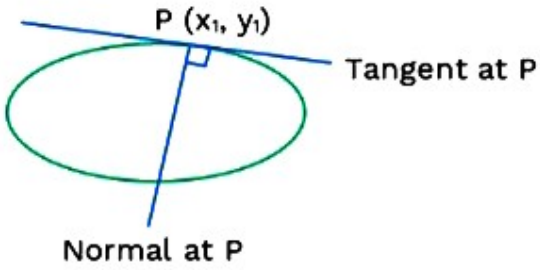
$$\frac{xx_1}{a^2}+\frac{yy_1}{b^2} = 1$$

The slope of the tangent at $\left(x_1, y_1\right) = \frac{-x_1}{a^2}\times\frac{b^2}{y_1}$

$\therefore$ Slope of the normal at $\left(x_1, y_1\right) = \frac{a^2}{x_1}\times\frac{y_1}{b^2} = \frac{a^2y_1}{b^2x_1}$

Hence the equation of the normal at $(x_1, y_1)$ is

$$y - y_1 = \frac{a^2y_1}{b^2x_1}\left(x - x_1\right)$$




### (ii) Parametric form:

In above equation if we put $x = a\cos\theta$ and $y = b\sin\theta$ then we will get normal equation in parametric form.

$\therefore$ ax $\sec\theta$-by $\operatorname{cosec}\theta = a^2-b^2 = a^2e^2$

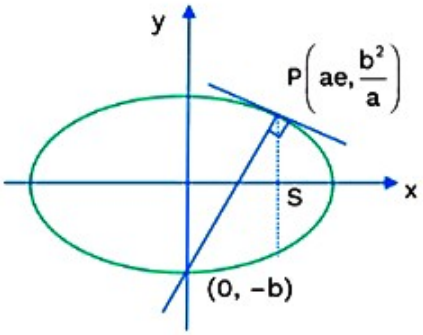
This is equation of normal in parametric form.

**Point to Remember!!!**

Equation of a normal in terms of its slope 'm' is $y = mx - \frac{\left(a^2-b^2\right)m}{\sqrt{a^2+b^2m^2}}$.

**Q.** **If the normal at one end of a latus-rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through other extremity of the minor axis, then show that $e^2=\frac{\sqrt{5}-1}{2}$.**

**Sol.** Equation of normal at $P\left(ae,\frac{b^2}{a}\right)$




is $\frac{x-ae}{\frac{ae}{a^2}}=\frac{y-\frac{b^2}{a}}{\frac{b^2/a}{b^2}}$

$\Rightarrow \frac{ax}{e}-ya=a^2-b^2$ Since it passes through (0,-b)

$\Rightarrow 0+ab=a^2-b^2 \Rightarrow a^2b^2=\left(a^2-b^2\right)^2$

$\Rightarrow a^2\cdot a^2\left(1-e^2\right)=\left(a^2\cdot e^2\right)^2 \Rightarrow e^4+e^2-1=0 \Rightarrow e^2=\frac{\sqrt{5}-1}{2}$.

**Q.** **Find the condition that the line lx + my = n may be a normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.**

**Sol.** Equation of normal to the given ellipse at $(a\cos\theta, b\sin\theta)$ is $\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2$

If the line lx + my = n is also normal to the ellipse then there must be a value of θ for which line (1) and line lx + my = n are identical. For that value of θ we have

$$\frac{l}{\left(\frac{a}{\cos\theta}\right)}=\frac{m}{\left(\frac{-b}{\sin\theta}\right)}=\frac{n}{\left(a^2-b^2\right)} \quad \text{or} \quad \frac{l}{a}\cos\theta=-\frac{m\sin\theta}{b}=\frac{n}{\left(a^2-b^2\right)}$$

$$\therefore \cos\theta=\frac{an}{l.\left(a^2-b^2\right)} \qquad ...(3)$$

$$\text{and } \sin\theta=\frac{-bn}{\left(a^2-b^2\right)m} \qquad ...(4)$$

Squaring and adding (3) and (4), we get $1=\frac{n^2}{\left(a^2-b^2\right)^2}\left(\frac{a^2}{l^2}+\frac{b^2}{m^2}\right)$ which is the required solution.

**Q.** **If normal at any point P to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\ (a>b)$ meet the axes at M and N so that $\frac{PM}{PN}=\frac{2}{3}$, then the value of eccentricity is:**

**(A)** $\frac{1}{\sqrt{2}}$ **(B)** $\frac{\sqrt{2}}{\sqrt{3}}$ **(C)** $\frac{1}{\sqrt{3}}$ **(D) None of these**

**Sol.** **(C)**

$(PM)^2 = a^2\cos^2\theta(1-e^2)^2 + b^2\sin^2\theta$

$= b^2(1-e^2\cos^2\theta)$

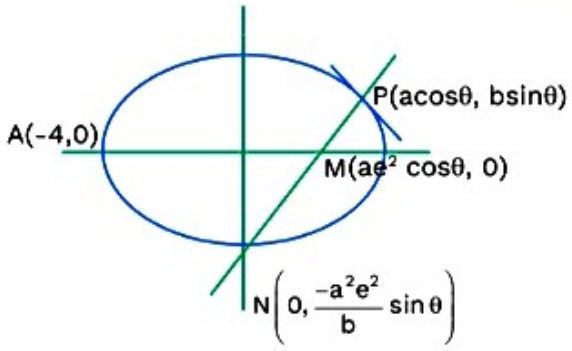
$(PN)^2 = a^2\cos^2\theta+\frac{\sin^2\theta}{b^2}\left(b^2+a^2e^2\right)^2$

$=\frac{a^4}{b^2}\left(1-e^2\cos^2\theta\right)$

$\left(\frac{PM}{PN}\right)^2=\frac{b^4}{a^4}=\frac{4}{9}$

$\frac{b^2}{a^2}=\frac{2}{3}=1-e^2$

$e=\frac{1}{\sqrt{3}}$




**Q.** **If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2+y^2+4x+1=0$, then find the maximum value of ab.**

**Sol.** **(4)**

Equation of tangent of ellipse at point (a cosθ, b sinθ) is $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$

It passes through (−2, 0)

$\Rightarrow -\frac{2}{a}\cos\theta=1$

Also its slope $=-\frac{b}{a}\cot\theta=2$

$\therefore \sec\theta=-\frac{2}{a},\tan\theta=-\frac{b}{2a}$

$\therefore \frac{4}{a^2}=1+\frac{b^2}{4a^2}$

$\Rightarrow 16 = 4a^2+b^2 \geq 2(2ab) \Rightarrow ab \leq 4$

**Q.** **If the normal at any point P of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the major and minor axes at G and E. respectively, and if CF is perpendicular upon this normal from the centre C of the ellipse, show that :PF · PG = $b^2$ and PF.PE = $a^2$**

**Sol.** Given ellipse is

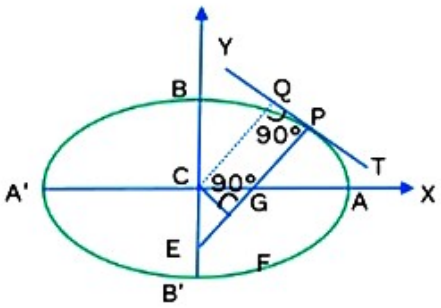
$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \qquad ...(1)$$

Let P(acosθ,bsinθ) be any point on ellipse (1)
Equation of normal to ellipse (1) at P is

$$ax\sec\theta - by\operatorname{cosec}\theta = a^2-b^2 \qquad ...(2)$$

since line (2) meets the major axis (x- axis) and minor axis (y-axis ) at G and E respectively, therefore

$$G \equiv \left(\frac{a^2-b^2}{a}\cos\theta, 0\right) \text{ and } E \equiv \left(0, \frac{-(a^2-b^2)}{b}\sin\theta\right)$$




PF = CQ = length of perp. form C(0,0) on the tangent at P,

i.e. on the line $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}-1=0$

$$= \frac{|-1|}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}} = \frac{ab}{\sqrt{b^2\cos^2\theta + a^2\sin^2\theta}} \qquad ...(3)$$

$$\text{Also PG} = \sqrt{\left\{\frac{(a^2-b^2)\cos\theta}{a} - a\cos\theta\right\}^2 + (0-b\sin\theta)^2}$$

$$= \sqrt{\frac{b^4\cos^2\theta}{a^2}+b^2\sin^2\theta} = \frac{b}{a}\sqrt{b^2\cos^2\theta + a^2\sin^2\theta} \qquad ...(4)$$

$$PE = \sqrt{a^2\cos^2\theta + \left(b\sin\theta + \frac{a^2-b^2}{b}\sin\theta\right)^2}$$

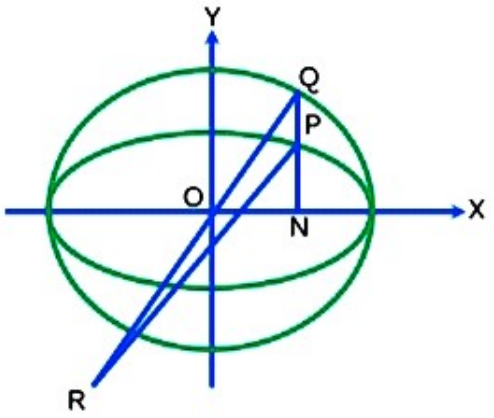
$$= \sqrt{a^2\cos^2\theta + \frac{a^4\sin^2\theta}{b^2}} = \frac{a}{b}\sqrt{b^2\cos^2\theta + a^2\sin^2\theta} \qquad ...(5)$$

From (3) and (4), PF.PG = $b^2$
From (3) and (5), PF.PE = $a^2$

**Q.** **Any ordinate NP of an ellipse meets the auxiliary circle in Q, prove that the locus of the intersection of the normals at P and Q is the circle $x^2 + y^2 = (a + b)^2$**

**Sol.** Equation of the ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$




Equation of its auxiliary circle is

$$x^2 + y^2 = a^2 \quad \ldots(2)$$

If coordinates of P are $(a\cos\phi, b\sin\phi)$, then those of Q are $(a\cos\phi, a\sin\phi)$

Equation of the normal to the ellipse at P is

$$ax\sec\phi - by\,\text{cosec}\,\phi = a^2 - b^2 \quad \ldots(3)$$

Equation of the normal to the circle at Q is

$$y = x\tan\phi \quad \ldots(4)$$

By eliminating $\phi$ from equations (3) and (4), we will get the required locus.

From the equation (4), we have

$$\sec\phi = \frac{\sqrt{x^2 + y^2}}{x} \text{ and } \text{cosec}\,\phi = \frac{\sqrt{x^2 + y^2}}{y}$$

Substituting in the equation (3), we get

$$a\sqrt{x^2 + y^2} - b\sqrt{x^2 + y^2} = a^2 - b^2$$

or $\sqrt{x^2 + y^2} = a + b$ or $x^2 + y^2 = (a + b)^2$

Which is the required locus.

**Q.** **If the normals at $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ to the ellipse are concurrent, the prove that**

$$\begin{vmatrix} x_1 & y_1 & x_1y_1 \\ x_2 & y_2 & x_2y_2 \\ x_3 & y_3 & x_3y_3 \end{vmatrix} = 0$$

**Sol.** The equations of the normals at $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ to the ellipse are

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$$

$$\frac{a^2x}{x_2} - \frac{b^2y}{y_2} = a^2 - b^2$$

$$\frac{a^2x}{x_3} - \frac{b^2y}{y_3} = a^2 - b^2, \text{ respectively}$$

These three lines will be concurrent, if

$$\begin{vmatrix} \frac{a^2}{x_1} & \frac{-b^2}{y_1} & a^2-b^2 \\ \frac{a^2}{x_2} & \frac{-b^2}{y_2} & a^2-b^2 \\ \frac{a^2}{x_3} & \frac{-b^2}{y_3} & a^2-b^2 \end{vmatrix} = 0$$

$$\Rightarrow -a^2b^2\left(a^2-b^2\right)\begin{vmatrix} \frac{1}{x_1} & \frac{1}{y_1} & 1 \\ \frac{1}{x_2} & \frac{1}{y_2} & 1 \\ \frac{1}{x_3} & \frac{1}{y_3} & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} y_1 & x_1 & x_1y_1 \\ y_2 & x_2 & x_2y_2 \\ y_3 & x_3 & x_3y_3 \end{vmatrix} = 0$$

**DIRECTOR CIRCLE:**

Locus of the point of intersection of the tangents which meet at right angles is called the Director Circle. Let equation of any tangent is $y = mx + \sqrt{a^2 m^2 + b^2}$ If it passes through (h, k) then

$$k = mh \pm \sqrt{a^2 m^2 + b^2}$$

$(k - mh)^2 = a^2m^2 + b^2$

$(h^2 - a^2)m^2 - 2khm + k^2 - b^2 = 0$ ...(3)

Equation (3) has two roots $m_1$ and $m_2$

$$m_1 + m_2 = \frac{2hk}{h^2 - a^2} \quad ...(4)$$

$$m_1 m_2 = \frac{k^2 - b^2}{h^2 - a^2} \quad ...(5)$$

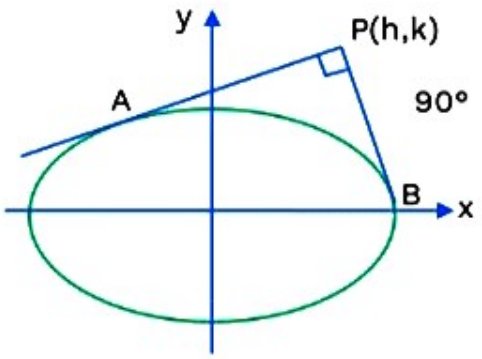
Hence passing through a given point there can be a maximum of two tangents.

If PA⊥PB then $m_1 m_2 = -1$

i.e. $m_1 m_2 = \frac{k^2 - b^2}{h^2 - a^2}$

i.e. $k^2 - b^2 = a^2 - h^2$ ;

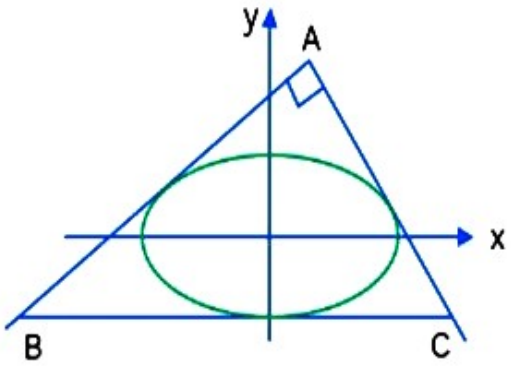
i.e. $x^2 + y^2 = a^2 + b^2$




**Point to Remember!!!**

If a right triangle, right angled at A circumscribes an ellipse then locus of the point A is the director circle of the ellipse.

which is the director circle of the ellipse. Hence director circle of an ellipse is a circle whose centre is the centre of ellipse and whose radius is the length of the line joining the ends of the major and minor axis.




Equation (3) can be used to determine the locus of the point of intersection of two tangents enclosing. If from any point P(h,k) pair of tangents are drawn to the ellipse which include an angle $\alpha$, then

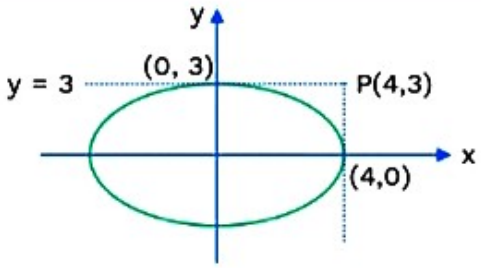
$$\tan\alpha = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2}$$

By putting value of $m_1 + m_2$ and $m_1 m_2$ in above equation we will get the angle between pair of tangents.

**Q.** **How many real tangents can be drawn from the point (4,3) to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. Find the equation of these tangents and angle between them.**

**Sol.** Given point P = (4,3)
ellipse

$$S = \frac{x^2}{16} + \frac{y^2}{9} - 1 = 0 \because S_1 = \frac{16}{16} + \frac{9}{9} - 1 = 1 > 0$$




$\Rightarrow$ Point P = (4, 3) lies outside on director circle of the ellipse.
$\therefore$ Two tangents can be drawn from the point P(4, 3). One x = 4 and other y = 3
angle between them $= \frac{\pi}{2}$

**Q.** **An ellipse slides between two lines at right angles to another. Show that the locus of its centre is a circle.**

**Sol.** Let the two given perpendicular lines be taken as the x and y axes respectively.
Let C($\alpha$, $\beta$) be the centre of the ellipse in any position.
Here the position of centre C changes as the ellipse slides.

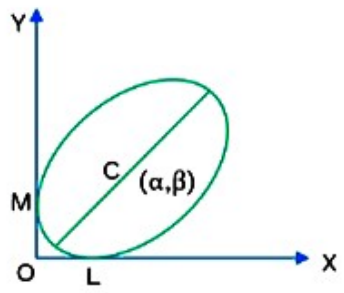
Let a and b be the semimajor and minor axes of the ellipse. Equation of the director circle of the ellipse is

$(x-\alpha)^2 + (y-\beta)^2 = a^2 + b^2$ ...(1)

Since OX and OY are mutually perpendicular tangents to sliding ellipse for all its positions, therefore, O(0,0) will lie on its director circle (1)

$\therefore \alpha^2 + \beta^2 = a^2 + b^2$ ...(1)

Hence locus of C(a, b) is $x^2 + y^2 = a^2 + b^2$ ...(2)




**Q.** **The length of sides of square which can be made by four perpendicular tangents to the ellipse $\frac{x^2}{7}+\frac{2y^2}{11}=1$ is :**

**(A) 1** **(B) 5** **(C) 6** **(D) None of these**

**Sol.** **(B)**

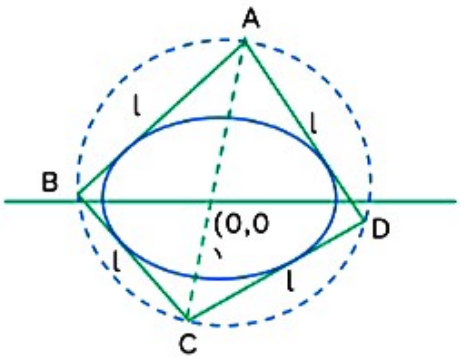
$$\ell^2 + \ell^2 = (AC)^2 = \left(2\sqrt{a^2+b^2}\right)^2$$

$$2l^2 = 4\left(7+\frac{11}{2}\right)$$

$l^2 = 14 + 11 = 25$

$l = 5$

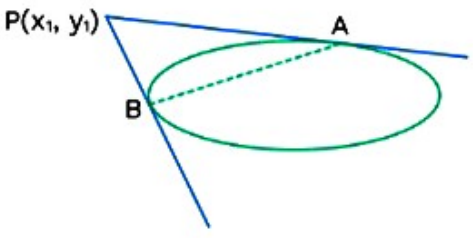
$$\Rightarrow \tan\theta = \frac{1}{\sqrt{2}} \quad \Rightarrow P \equiv \left(\sqrt{6}\frac{\sqrt{2}}{\sqrt{3}}, \sqrt{3}\frac{1}{\sqrt{3}}\right) \equiv (2,1)$$




**CHORD OF CONTACT:**

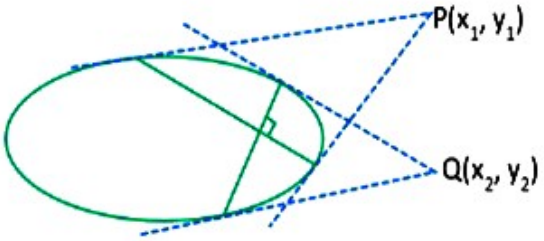
Pair of tangents drawn from outside point $P(x_1, y_1)$ to the ellipse which meet it at A and B. Now line joining A and B is called the chord of contact of point $P(x_1, y_1)$ w.r.t. the ellipse. The equation of chord of contact is

$$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$$



**Q.** **If the chords of contact of tangents from two points $(x_1, y_1)$ and $(x_2, y_2)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are at right angles, then Find the value of $\frac{x_1x_2}{y_1y_2}$.**

**Sol.** The equations of the chords of contact of tangents drawn from $(x_1, y_1)$ and $(x_2, y_2)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are



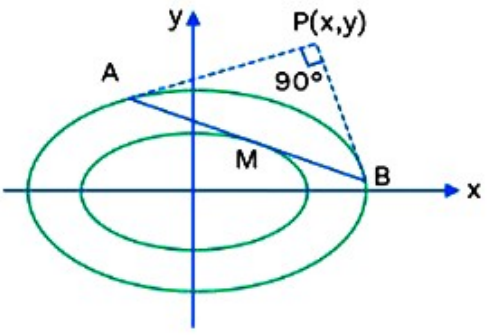

$$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1 \;\&\; \frac{xx_2}{a^2}+\frac{yy_2}{b^2}=1$$

It is given that (i) and (ii) are at right angles.

$$\therefore \frac{-b^2}{a^2}\frac{x_1}{y_1}\times\frac{-b^2}{a^2}\frac{x_2}{y_2}=-1 \Rightarrow \frac{x_1x_2}{y_1y_2}=-\frac{a^4}{b^4}$$

**Q.** **Show that the tangents drawn at those points of the ellipse $\frac{x^2}{a}+\frac{y^2}{b}=(a+b)$, where it is cut by any tangent to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, intersect at right angles.**

**Sol.** Given ellipse are $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ...(i)




and

$$\frac{x^2}{a(a+b)}+\frac{y^2}{b(a+b)}=1 \quad ...(ii)$$

The chord of contact $(x_1, y_1)$ w.r.t. (ii) ellipse is

$$\frac{xx_1}{a(a+b)}+\frac{yy_1}{b(a+b)}=1$$

i.e. $y=\frac{-bx_1}{ay_1}x+\frac{b(a+b)}{y_1}$ ...(iii)

(iii) is tangent to ellipse (i) which is also given as

$y=mx+\sqrt{a^2m^2+b^2}$ ...(iv)

Hence, (iii) & (iv) is identical

$\therefore\ m = -\dfrac{bx_1}{ay_1}$ ...(v)

and $a^2m^2 + b^2 = \dfrac{b^2(a+b)^2}{y_1^2}$ ...(vi)

Now eliminate m from (v) & (vi), we get

$x_1^2 + y_1^2 = (a+b)^2$

$\therefore$ Locus of $P(x,y)$ is $x^2 + y^2 = (a+b)^2$

**Q.** **Equation of the chord of contact of pair of tangents drawn to the ellipse $4x^2 + 9y^2 = 36$ from the the point (m, n) where m,n = m + n, m, n being non-zero positive integer is**

**(A) $2x + 9y = 18$ (B) $2x + 2y = 1$ (C) $4x + 9y = 18$ (D) None of these**

**Sol.** **(C)**

Given $m(n-1) = n$

n is divisible by $n - 1$

$\Rightarrow n = 2 \Rightarrow m = 2$

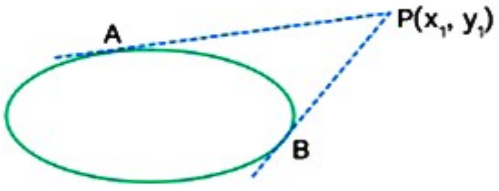
Hence, chord of contanct of tangents drawn from (2, 2) to $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ is

$\dfrac{2x}{9} + \dfrac{2y}{4} = 1$

$\Rightarrow 4x + 9y = 18$

**PAIR OF TANGENTS:**

Pair of tangents PA and PB are drawn from outside point $P(x_1,y_1)$, which is shown below. Hence joint equation of line PA and PB is given by $SS_1 = T^2$.




Here $S \equiv \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - 1$

$S_1 \equiv \dfrac{x_1^2}{a^2} + \dfrac{y_1^2}{b^2} - 1$

$T \equiv \dfrac{xx_1}{a^2} + \dfrac{yy_1}{b^2} - 1$

**Q.** **Find the locus of point of intersection of perpendicular tangents to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$**

**Sol.** Let P(h, k) be the point of intersection of two perpendicular tangents

$$\Rightarrow \left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}-1\right)=\left(\frac{hx}{a^2}+\frac{ky}{b^2}-1\right)^2$$

$$\Rightarrow \frac{x^2}{a^2}\left(\frac{k^2}{b^2}-1\right)+\frac{y^2}{b^2}\left(\frac{h^2}{a^2}-1\right)+\ldots\ldots\ldots=0 \qquad \ldots(i)$$

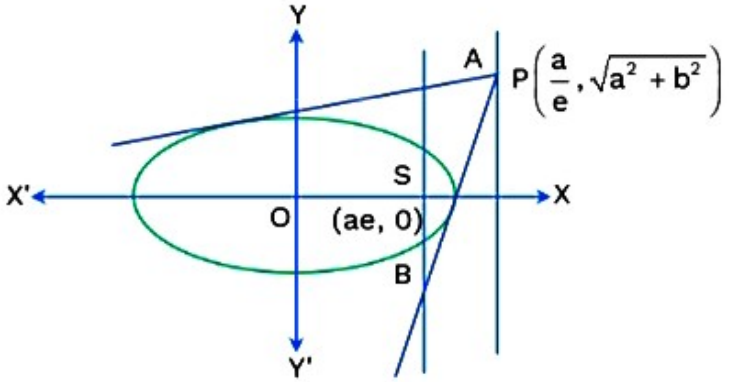
Since equation (i) represents two perpendicular lines

$$\therefore \frac{1}{a^2}\left(\frac{k^2}{b^2}-1\right)+\frac{1}{b^2}\left(\frac{h^2}{a^2}-1\right)=0$$

$$\Rightarrow k^2-b^2+h^2-a^2=0 \Rightarrow \text{ locus is } x^2+y^2=a^2+b^2$$

**Q.** **Tangent area drawn to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from the point $\left(\frac{a^2}{\sqrt{a^2-b^2}},\sqrt{a^2+b^2}\right)$, Prove that they intercept on the ordinate through the nearer focus a distance equal to the major axis.**

**Sol.** Equation of pair of tangents from point P is $SS_1 = T^2$




$$\Rightarrow \left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)\left(\frac{1}{e^2}+\frac{a^2+b^2}{b^2}-1\right)=\left(\frac{xa}{ea^2}+\frac{y\sqrt{a^2+b^2}}{b^2}-1\right)^2$$

Solving this with x = ae, we get

$$\left(e^2+\frac{y^2}{b^2}-1\right)\left(\frac{1}{e^2}+\frac{a^2}{b^2}\right)=\frac{y^2\left(a^2+b^2\right)}{b^2b^2}$$

$$\Rightarrow \left(\frac{y^2}{b^2}-\frac{b^2}{a^2}\right)\left(\frac{b^2+a^2-b^2}{b^2e^2}\right)=\frac{y^2\left(a^2+b^2\right)}{b^2b^2}$$

$$\left(\text{using } e^2=1-\frac{b^2}{a^2}\right)$$

$$\Rightarrow \left(\frac{a^2y^2-b^4}{a^2b^2}\right)\left(\frac{a^2}{b^2e^2}\right)=\frac{y^2\left(a^2+b^2\right)}{b^4}$$

$$\Rightarrow \quad a^2\left(a^2y^2-b^4\right)=y^2\left(a^4-b^4\right)$$

[using $(a^2e^2 = a^2 - b^2)$]

$\Rightarrow a^2b^4 = y^2b^4$

$\Rightarrow y^2 = a^2$

$\Rightarrow y = \pm a \Rightarrow AB = 2a$

**Q.** **The set of values of m for which it is possible to draw the chor $y=\sqrt{m}x+1$ to the curve $x^2 + 2xy + (2 + \sin^2\alpha)y^2 = 1$, which subtends a right angle at the origin for some value of $\alpha$ is**

**(A) [2,3]** **(B) [0,1]** **(C) [1,3]** **(D)None of these**

**Sol.** **(A)**

Combined equation of pair of lines through the origin joining the points of intersection of line $y=\sqrt{m}x+1$ with the given curve is

$x^2 + 2xy + (2 + \sin^2\theta)y^2 - (y-\sqrt{m}x)^2 = 0$

For the chord to subtend a right angle at the origin $(1 - m) + (2 + \sin^2\alpha - 1) = 0$ (as sum of the coefficients of $x^2 + y^2 = 0$)

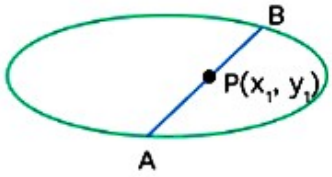
$\Rightarrow \sin^2\alpha = m - 2$

$\Rightarrow 0 \le m - 2 \le 1$

$\Rightarrow 2 \le m \le 3$

## CHORD WITH A GIVEN MIDDLE POINT:

Here chord AB is shown in the figure whose mid point is $P(x_1, y_1)$.




Then equation of this chord AB is $T = S_1$ Here

$$S_1 \equiv \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$$

$$T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$$

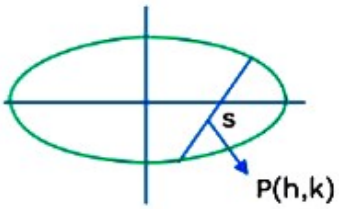
**Q.** **Find the locus of the mid - point of focal chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.**

**Sol.** Let P=(h,k) be the mid-point

$\therefore$ Equation of chord whose mid-point is given $\frac{xh}{a^2} + \frac{yk}{b^2} - 1 = \frac{h^2}{a^2} + \frac{k^2}{b^2} - 1$

Since it is a focal chord, therefore it passes through focus, either (ae, 0) or (−ae, 0) If it passes through (ae, 0)

$\therefore$ locus is $\frac{ex}{a} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$




If it passes through $(-ae, 0)$

$\therefore$ locus is $-\frac{ex}{a} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Ans.

**Q.** **Tangents are drawn from the point (3,2) to the ellipse $x^2 + 4y^2 = 9$. Find the equation to their chord of contact and the middle point of this chord of contact.**

**Sol.** $x^2 + 4y^2 = 9$

Equation of the chord of contact of the pair of tangents from (3, 2)

$3x + 8y = 9$ ...(i)

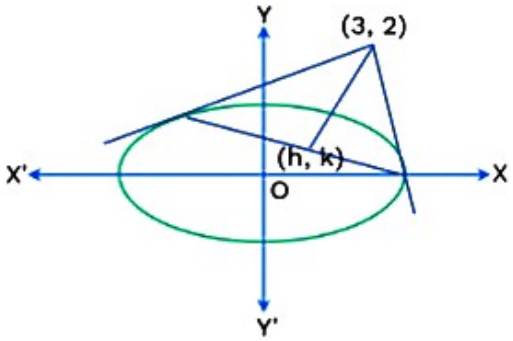
This must be the same as chord whose middle point is (h, k)

$T = S_1$

$$\frac{hx}{9} + \frac{ky}{9/4} = \frac{h^2}{9} + \frac{k^2}{9/4}$$

$\Rightarrow hx + 4ky = h^2 + 4k^2$ ...(ii)

Equations (i) and (ii) represent same straight lines.



Comparing coefficient of (i) and (ii),
we get

$$\frac{h}{3} = \frac{4k}{8} = \frac{h^2 + 4k^2}{9}$$

$\Rightarrow$ $2h = 3k$ and $3h = h^2 + 4k^2$

$$\Rightarrow \quad 3h = h^2 + 4 \times \frac{4h^2}{9}$$

$$\Rightarrow \quad \frac{25h^2}{9} = 3h$$

$$\Rightarrow \quad h = \frac{27}{25} \text{ and } k = \frac{18}{25}$$

**Q.** **The normal at a variable point P on an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e' such that :**

**(A) e' is independent off e** **(B) e' = 1**

**(C) e' = e** **(D) e' = 1/e**

**Sol.** **(C)**

$2h = ae^2 q$

$$2k = \frac{-a^2e^2 \sin\theta}{b}$$

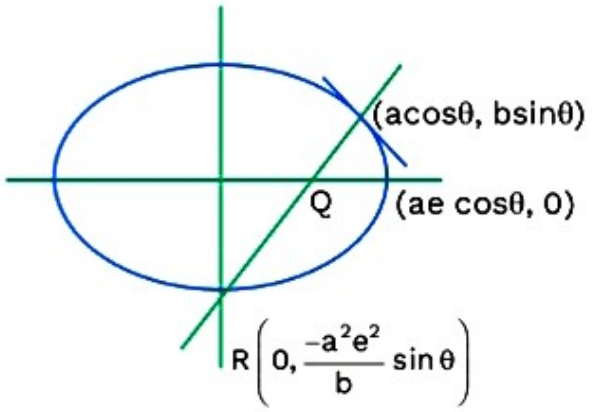
$$\Rightarrow \frac{4h^2}{\left(ae^2\right)^2} + \frac{4k^2}{\left(a^2\frac{e^2}{b}\right)^2} = 1$$

$$\frac{x^2}{\left(\frac{ae^2}{2}\right)^2} + \frac{y^2}{\left(\frac{a^2e^2}{2b}\right)^2} = 1$$

$$\frac{a^2e^4}{4} = \frac{a^4e^4}{4b^2}\left(1-(e')^2\right)$$

$b^2 = a^2(1 - (e')^2)$

$\Rightarrow$ $e = e'$



**Q.** **Find the locus of the middle points of chords of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, the tangents at the ends of which intersect at right angles.**

**Sol.** Let M(h, k) be the mid-point of a chord QR, then its equation is

$$T = S_1$$

$$\Rightarrow \quad \frac{hx}{a^2}+\frac{ky}{b^2}=\frac{h^2}{a^2}+\frac{k^2}{b^2} \qquad ...(1)$$

If the tangents at the ends Q and R of this chord meet perpendicularly in $P(x_1, y_1)$, then P must lie on the director circle, whose equation is $x^2 + y^2 = a^2 + b^2$. So,

$$x_1^2+y_1^2=a^2+b^2 \qquad ...(2)$$

Now the equation of the chord of contact of the tangents drawn from P is

$$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1 \qquad ...(3)$$

Clearly the equations (1) and (3) represent the same line.
Thus,

$$\frac{x_1}{h}=\frac{y_1}{k}=\frac{1}{\frac{h^2}{a^2}+\frac{k^2}{b^2}}$$

$$\Rightarrow \quad x_1=\frac{h}{\frac{h^2}{a^2}+\frac{k^2}{b^2}} \text{ and } y_1=\frac{k}{\frac{h^2}{a^2}+\frac{k^2}{b^2}}$$

Substituting the values of $x_1$ and $y_1$ in the equation (2), we get

$$h^2+k^2=\left(a^2+b^2\right)\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2$$

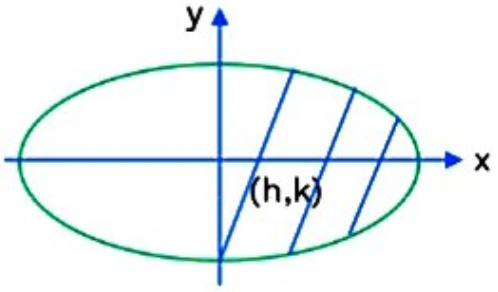
Hence, the required locus of M (h, k) is

$$x^2+y^2=\left(a^2+b^2\right)\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)^2$$

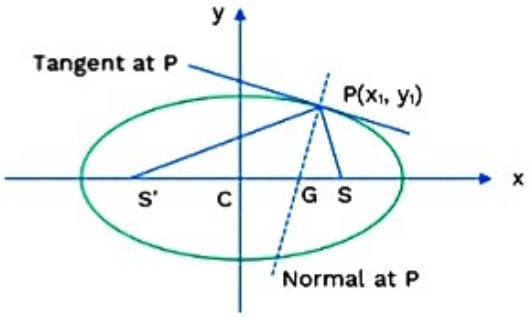
**DIAMETER :**

The locus of the middle points of a system of parallel chords with slope 'm' of an ellipse is a straight line passing through the centre of the ellipse, called its diameter and has the equation

$$y=-\frac{b^2}{a^2 m}x$$



Chord of contact, pair of tangents, chord with a given middle point are to be interpreted as same as they are in parabola.




**Properties of the ellipse:**

**(1)** Let the equation of an ellipse is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \qquad ...(i)$$

& Normal at $P(x_1,y_1)$ is $\frac{x-x_1}{x_1/a^2}=\frac{y-y_1}{y_1/b^2}$ ...(ii)

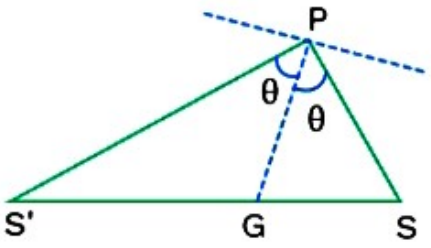
The normal meet x -axis at G ⇒ Put y = 0 in equation (ii) we get

$$CG = x = \left(\frac{a^2-b^2}{a^2}\right)x_1 = e^2x_1$$

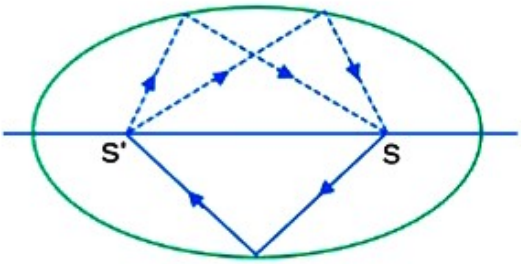
$$\therefore\ SG = CS - CG = ae - e^2x_1 = e(a - ex_1) = eSP$$

Similarly S' G = eSP

$$\therefore\ \frac{SG}{S'G}=\frac{eSP}{eS'P}=\frac{SP}{S'P} \Rightarrow$$ PG is bisector of angle ∠P in ΔS' PS.




Tangent and normal at any point P bisect the external & internal angles between the focal distances of SP and S'P.



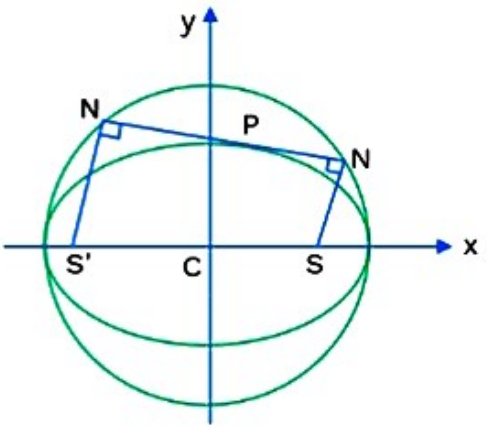

This lead to reflection property of ellipse.
If incoming light ray passes through focus S' (or S), strike the concave side of ellipse the after reflection it will pass through other focus.

**(2)** Let ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ...(i)

its tangent is $y - mx = \sqrt{a^2m^2+b^2}$ ...(ii)

Equation of perpendicular to above the passes through focus (ae, 0) is

my + x = ae ...(iii)




Eliminate m from (ii) & (iii) we will get focus of intersection point.
For that square & add (ii) & (iii) we will get an answer

$\therefore\ y^2 + m^2x^2 - 2mxy + x^2 + my^2 + 2mxy = a^2m^2 + b^2 + a^2e^2$

$\Rightarrow\ x^2\left(1+m^2\right) + y^2\left(1+m^2\right) = a^2m^2 + a^2$

$\Rightarrow\ x^2 + y^2 = a^2$ which is the auxiliary circle.

The locus of the point of intersection of feet of perpendicular from foci on any tangent to an ellipse is the auxiliary circle.

**(3)** From previous equation of any tangent is

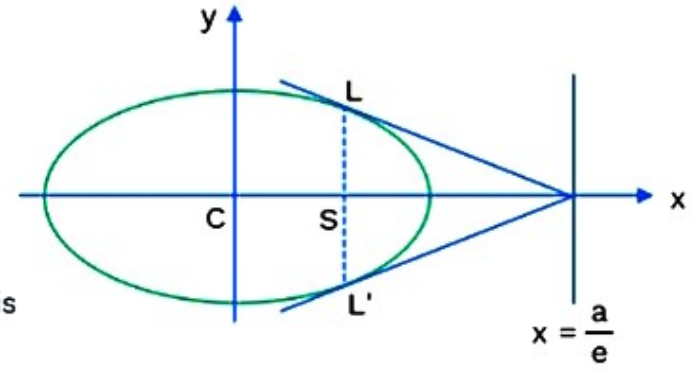
$mx - y + \sqrt{a^2m^2 + b^2} = 0$

$$SN = \left|\frac{\sqrt{a^2\ m^2 + b^2} + am}{\sqrt{1+m^2}}\right| \ \&\ \ S'N' = \left|\frac{\sqrt{a^2\ m^2 + b^2} - ame}{\sqrt{1+m^2}}\right|$$

$$SN \cdot S'N' = \frac{\left(a^2\ m^2 + b^2\right) - a^2\ m^2e^2}{\left(1+m^2\right)} = \frac{a^2\ m^2 + b^2 - \left(a^2 - b^2\right)m^2}{1+m^2} = b^2 \because a^2e^2 = a^2 - b^2$$

The product of perpendicular distance from the foci to any tangent of an ellipse is equal to square of the semi minor axis.

**(4)** Equation of tangent at $L\left(ae, \frac{b^2}{a}\right)$ is




$$\frac{x(ae)}{a^2} + \frac{y\left(\frac{b^2}{a}\right)}{b^2} = 1 \Rightarrow xe + y = a \quad ...(i)$$

and equation of tangent at $L\left(ae, -\frac{b^2}{a}\right)$ is

$ex - y = a$ ...(ii)

Solve (i) & (ii) we get $x = \frac{a}{e}$ & $y = 0$ i.e. at the directrix of ellipse.

Tangents at the extremities of latus-rectum of an ellipse intersect on the corresponding directrix.

**(5)** If P be any point on the ellipse with S and S' as its foci then $l(SP) + l(S'P) = 2a$.

**(6)** If the normal at any point P on the ellipse with centre C meet the major & minor axes in G & g respectively & if CF be perpendicular upon this normal then

(i) $PF.\ PG = b^2$ (ii) $PF.Pg = a^2$

**(7)** The portion of the tangent to an ellipse between the point of contact & the directrix subtends a right angle at the corresponding focus.

**Q.** **The co-ordinates (2, 3) and (1, 5) are the foci of an ellipse which passes through the origin, then the equation of**

**(A) tangent at the origin is** $(3\sqrt{2}-5)x+(1-2\sqrt{2})y=0$

**(B) tangent at the origin is** $(3\sqrt{2}+5)x+(1+2\sqrt{2}y)=0$

**(C) normal at the origin is** $(3\sqrt{2}-5)x-(2\sqrt{2}+1)y=0$

**(D) normal at the origin is**

**Sol.** **(AC)**

Tangent and normal are bisectors of $\angle SPS'$

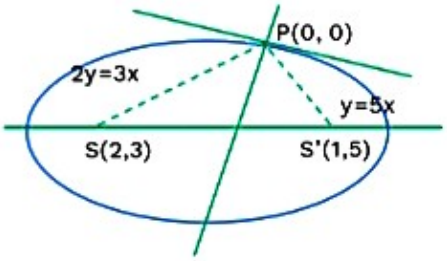
Now equation of SP is $y = 3x/2$ and that of $S'P$ is $y = 5x$

Then their equations are $\frac{3x-2y}{\sqrt{13}}=\pm\frac{5x-y}{\sqrt{26}}$

or $3x-2y=\pm\frac{5x-y}{\sqrt{2}}$

$\Rightarrow$ lines are $(3\sqrt{2}-5)x+(1-2\sqrt{2})y=0$

and $(3\sqrt{2}+5)x-(2\sqrt{2}+1)y=0$




Now (2, 3) and (1, 5) lie on the same side of $(3\sqrt{2}-5)x+(1-2\sqrt{2})y=0$, which is equation of tangent

Point (2, 3) and (1, 5) lie on the different side of $(3\sqrt{2}+5)x-(2\sqrt{2}+1)y=0$, which is equation of normal.

**Q.** **An ellipse has foci at (9, 20) and (49, 55) in the x-y plane and is tangent to x-axis. Let the length of its major axis is L, then** $\left[\frac{L}{11}\right]$ **= (where [·] = G.I.F)**

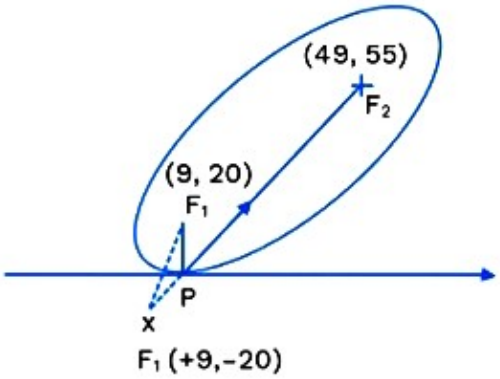
**Sol.** **(7)**

$2a = PF_1 + PF_2 = PF_1 + PF_2$

$= F_1F_2 = \sqrt{40^2+(75)^2} = 85$

$\Rightarrow \quad L = 85$

$\therefore \quad \left[\frac{L}{11}\right]=7$



**Comprehension for Q.1 and Q.2**

Let an ellipse having major axis and minor axis parallel to x-axis and y-axis respectively. Its two foci S and S' are (2, 1), (4, 1) and a line $x + y = 9$ is a tangent to this ellipse at point P.

**Q.1** **Eccentricity of the ellipse is :**

**(A)** $\frac{1}{2\sqrt{3}}$ **(B)** $\frac{1}{\sqrt{13}}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{1}{4}$

**Ans.** **(B)**

**Q.2** **Length of major axis is :**

**(A)** $\sqrt{13}$ **(B)** $2\sqrt{11}$ **(C)** $2\sqrt{13}$ **(D)** $4\sqrt{3}$

**Sol.** **(C)**

$2ae = 2 \Rightarrow ae = 1$ ...(1)

$P_1P_2 = b^2 = \frac{4}{\sqrt{2}}\frac{6}{\sqrt{2}} = 12$

$b^2 = 12$ ...(2)

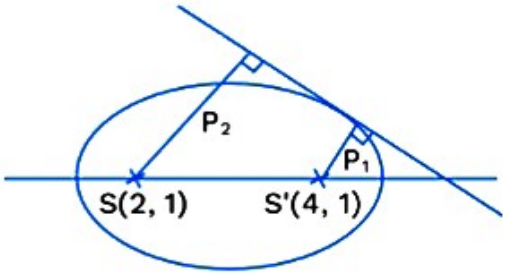
$b^2 = a^2 - a^2e^2 = a^2 - 1$

$\Rightarrow 12 = a^2 - 1 \Rightarrow a = \sqrt{13}$

$\Rightarrow$ Length of major axis $= 2a = 2\sqrt{13} = \sqrt{52}$

$ae = 1$ (from ...(1))

$\Rightarrow e = \frac{1}{\sqrt{13}}$




**Comprehension for Q.3 to Q.4**

A tangent is drawn at any point P $(4\cos\theta, 3\sin\theta)$ on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and on it is taken a point $Q(\alpha, \beta)$ from which pair of tangents QA and QB are drawn to the circle $x^2 + y^2 = 12$.

**Q.3** **The locus of the point of concurrency of chord of contact AB of the circle $x^2 + y^2 = 12$ is**

**(A)** $\frac{x^2}{9} + \frac{y^2}{16} = 1$ **(B)** $\frac{x^2}{12} + \frac{y^2}{4} = 1$ **(C)** $\frac{x^2}{12} + \frac{y^2}{16} = 1$ **(D)** $\frac{x^2}{6} + \frac{y^2}{8} = 1$

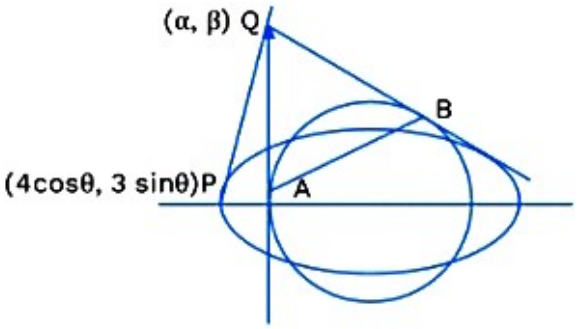
**Ans.** **(A)**

Equation of chord of contact AB is

$\alpha x + \beta y = 12$ ...(1)

Equation of tangent to ellipse at P is

$$\frac{x\cos\theta}{4} + \frac{y\sin\theta}{3} = 1$$




$\because$ $(\alpha, \beta)$ lies on it

$\Rightarrow$ $3\alpha\cos\theta + 4\beta\sin\theta = 12$ ...(2)

From eqn. (1) and (2)

Lines $\alpha x + \beta y = 12$ are concurrent at points $(3\cos\theta, 4\sin\theta)$

$\therefore$ Locus of point of concurrency is $\frac{x^2}{9} + \frac{y^2}{16} = 1$.

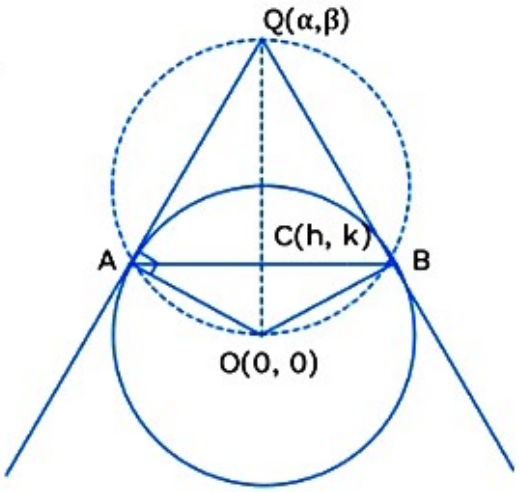
**Q.4** **The locus of the circumcentre of $\Delta QAB$ if $\theta = \frac{\pi}{4}$ is**

**(A)** $\frac{x}{4} + \frac{y}{3} = \sqrt{3}$ **(B)** $\frac{x}{4} + \frac{y}{3} = \frac{1}{\sqrt{2}}$ **(C)** $x^2 + y^2 = 6$ **(D)** $\frac{x}{3} + \frac{y}{4} = \frac{1}{\sqrt{2}}$

**Sol.** **(B)**

Equation of tangent to ellipse at $P\left(\frac{4}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$ is,

$$\frac{x}{4\sqrt{2}} + \frac{y}{3\sqrt{2}} = 1$$




$\because$ $(\alpha, \beta)$ lies on it

$\Rightarrow$ $3\alpha + 4\beta = 12\sqrt{2}$ ...(1)

From figure it is clear that OQ is diameter of circumcircle of $\Delta QAB$

$\Rightarrow$ $2h = \alpha$, $2k = \beta$

$\therefore$ From (1) we get

$3(2h) + 4(2k) = 12\sqrt{2}$

$\therefore$ The required. locus is $3x + 4y = 6\sqrt{2}$

**Q.** **Consider an ellipse $S_1 = \frac{x^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1$. Let $S_2 = 0$ be a parabola on the right of the y-axis confocal with $S_1 = 0$ having vertex at the centre of $S_1 = 0$. Let P be the point of intersection of the parabola and the directrix of ellipse in the first quadrant and L = 0 is the directrix of the parabola.**
**If the tangent of point Q on $S_1 = 0$ and the line joining the point P to the focus of $S_1 = 0$ intersect at the auxiliary circle of $S_1 = 0$, then eccentric angle of the**

**point Q (Q lies in first quadrant) is**

**(A)** $\tan^{-1}\left(\frac{e}{\sqrt{1+e^2}}\right)$ **(B)** $\tan^{-1}\left(\frac{2e}{\sqrt{1-e^2}}\right)$

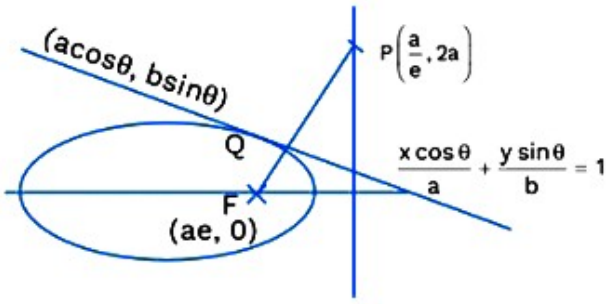
**(C)** $\tan^{-1}\left(\frac{\sqrt{2}e}{\sqrt{1+e^2}}\right)$ **(D)** $\tan^{-1}\left(\frac{e}{\sqrt{1-e^2}}\right)$

**Sol.** **(B)**

$\therefore$ Foot of perpendicular from focus of ellipse to any tangent to it lies on auxiliary circle.

$\Rightarrow$ FP $\perp$ Tangent at Q

$$\frac{2a-0}{\frac{a}{e}-ae}=\frac{a\sin\theta}{b\cos\theta}=\frac{a}{b}\tan\theta$$




$$\Rightarrow \tan\theta=\frac{b}{a}\frac{2e}{1-e^2}$$

$$=\frac{a\left(\sqrt{1-e^2}\right)2e}{a\left(1-e^2\right)}$$

$$\Rightarrow \theta=\tan^{-1}\left(\frac{2e}{\sqrt{1-e^2}}\right)$$

**Q.** **Let $P_1$ and P' be the feet of the perpendicular drawn foci foci S, S on a tangent T to an ellipse whose length of semi-major axis is 20, If $\sum_{i=1}^{10}(SP)(SP_i)=2560$, then the value of eccentricity is**

**(A)** $\frac{1}{5}$ **(B)** $\frac{2}{5}$ **(C)** $\frac{3}{5}$ **(D)** $\frac{4}{5}$

**Sol.** **(C)**

$$\sum_{i=1}^{10}(SP)(SP')=2560$$

$\Rightarrow 10b^2 = 2560$

$\Rightarrow b^2 = 256$

$\Rightarrow b = 16$

$\Rightarrow 256 = 400\ (1 - e^2) \Rightarrow 1-e^2=\frac{16}{25}$

$$\Rightarrow e=\frac{3}{5}$$

## Equation of an ellipse referred to two perpendicular lines :

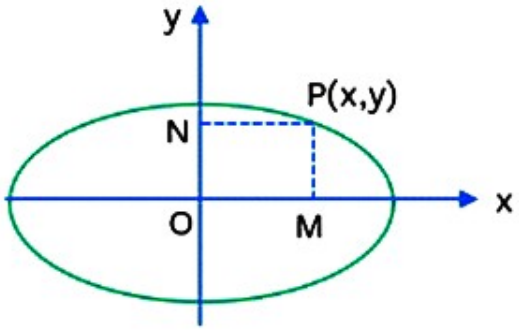
$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is given ellipse

Let P(x, y) be any point on the ellipse, then

PM = y and PN=x

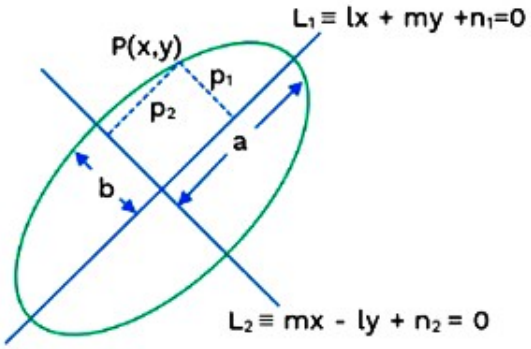
∴ above equation can be written as

$\frac{PN^2}{a^2}+\frac{PM^2}{b^2}=1$




From above we conclude that if perpendicular distances $p_1$ & $p_2$ of a moving point P(x, y) from two mutually perpendicular straight lines $L_1 \equiv lx + my + n_1 = 0$ & $L_2 \equiv mx - ly + n_2 = 0$ respectively then equation of ellipse in the plane of line will be $\frac{p_2^2}{a^2}+\frac{p_1^2}{b^2}=1$

$$\Rightarrow \frac{\left(\frac{mx-ly+n_2}{\sqrt{l^2+m^2}}\right)^2}{a^2}+\frac{\left(\frac{lx+my+n_1}{\sqrt{l^2+m^2}}\right)^2}{b^2}=1$$




**Q.** **Find the equation the ellipse whose axis are of length 6 & $2\sqrt{6}$ and their equations are x–3y + 3 = 0 and 3x + y – 1 = 0**

**Sol.** Equation of ellipse will be

$$\frac{\left(\frac{x-3x+3}{\sqrt{10}}\right)^2}{(\sqrt{6})^2}+\frac{\left(\frac{3x+y-1}{\sqrt{10}}\right)^2}{(3)^2}=1$$

**Q.** **Find the equation of the straight lines joining the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ to the foci of the ellipse $\frac{x^2}{24}+\frac{y^2}{49}=1$. Also find the area of the figure formed by the foci of these two ellipse.**

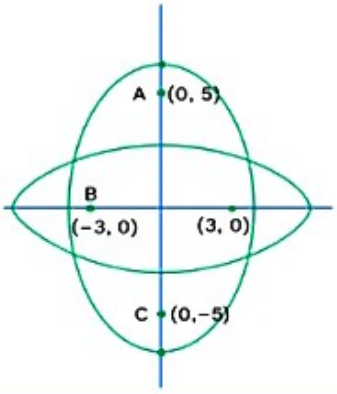
**Sol.** Foci of 1st ellipse (± 3, 0) foci of 2nd ellipse (0, ± 5)

equation of st. lines $\pm\frac{x}{3}\pm\frac{y}{5}=1$

Length of AB = $\sqrt{9+25}=\sqrt{34}$

Length of BC = $\sqrt{9+25}=\sqrt{34}$

Area = 34 sq. units




**Q.** **If Parametric point $\alpha, \beta, \gamma, \delta$ on ellipse are concyclic P.T. $\alpha, \beta, \gamma, \delta = 2n\pi$, $n \in I$.**

**Sol.** Let the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ cuts the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in four points P, Q, R, S.

Solving circle and ellipse ($x = a\cos\theta$, $y = b\sin\theta$), we have

$a^2\cos^2\theta + b^2\sin^2\theta + 2ag\cos\theta + 2bf\sin\theta + c = 0$

$$\Rightarrow a^2\left(\frac{1-t^2}{1+t^2}\right)^2 + b^2\left(\frac{2t}{1+t^2}\right)^2 + 2ag\left(\frac{1-t^2}{1+t^2}\right) + 2bf\left(\frac{2t}{1+t^2}\right) + c = 0, \text{ where } t = \tan\frac{\theta}{2}$$

$$\Rightarrow a^2\left(1-t^2\right)^2 + 4b^2t^2 + 2ag\left(1-t^2\right)\left(1+t^2\right) + 4bft\left(1+t^2\right) + c(1+t^2)^2 = 0$$

$$\Rightarrow \left(a^2 - 2ag + c\right)t^4 + 4bft^3 + \left(-2a^2 + 4b^2 + 2c\right)t^2 + 4bft + (a^2 + 2ag + c) = 0$$

Roots of the equation are $\tan\frac{\alpha}{2}, \tan\frac{\beta}{2}, \tan\frac{\gamma}{2}\tan\frac{\delta}{2}$, where $\alpha, \beta, \gamma, \delta$ are eccentric angles of P, Q, R, S, respectively.

Also

$$\Rightarrow \quad \tan\left(\frac{\alpha}{2}+\frac{\beta}{2}+\frac{\gamma}{2}+\frac{\delta}{2}\right) = \frac{s_1 - s_3}{1 - s_2 + s_4} = 0$$

$$\Rightarrow \quad \frac{\alpha+\beta+\gamma+\delta}{2} = n\pi, n \in Z$$

$$\Rightarrow \quad \alpha + \beta + \gamma + \delta = 2n\pi, n \in Z$$